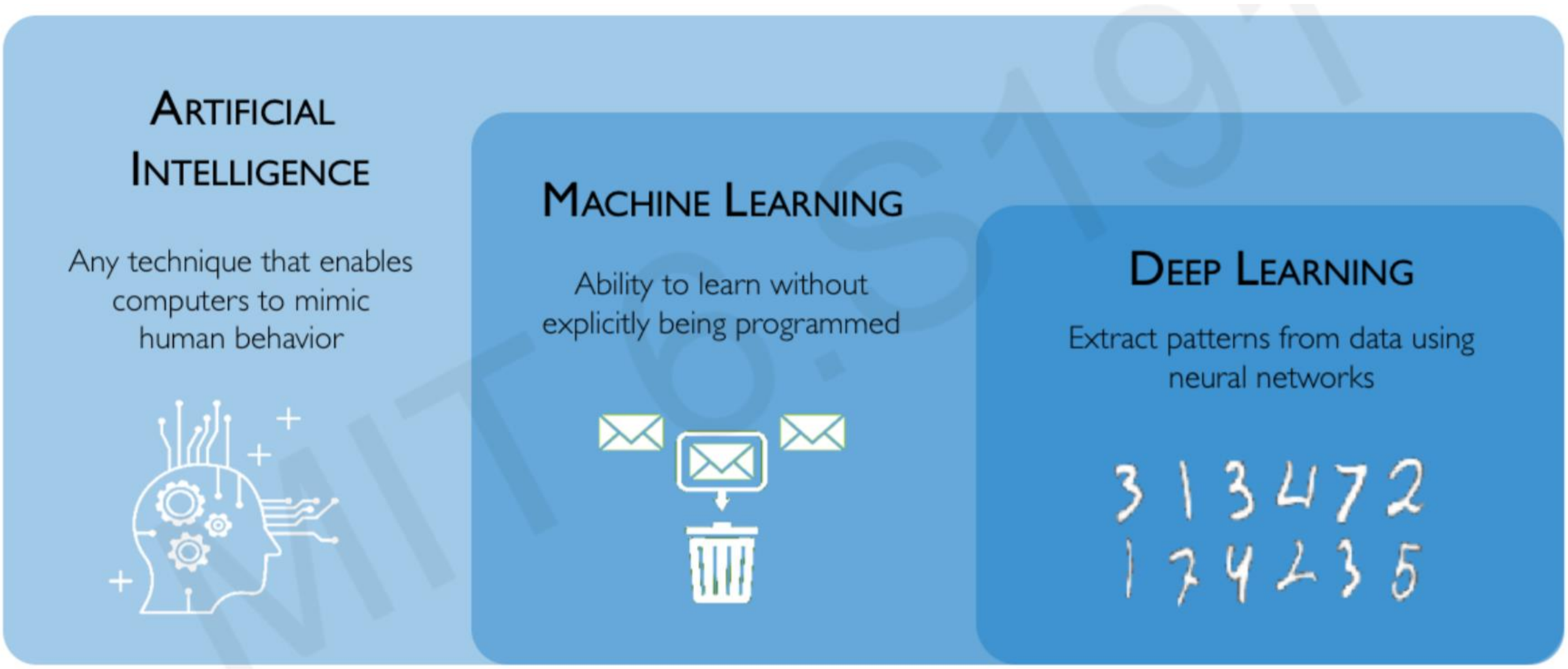
Introduction

What is deep learning?

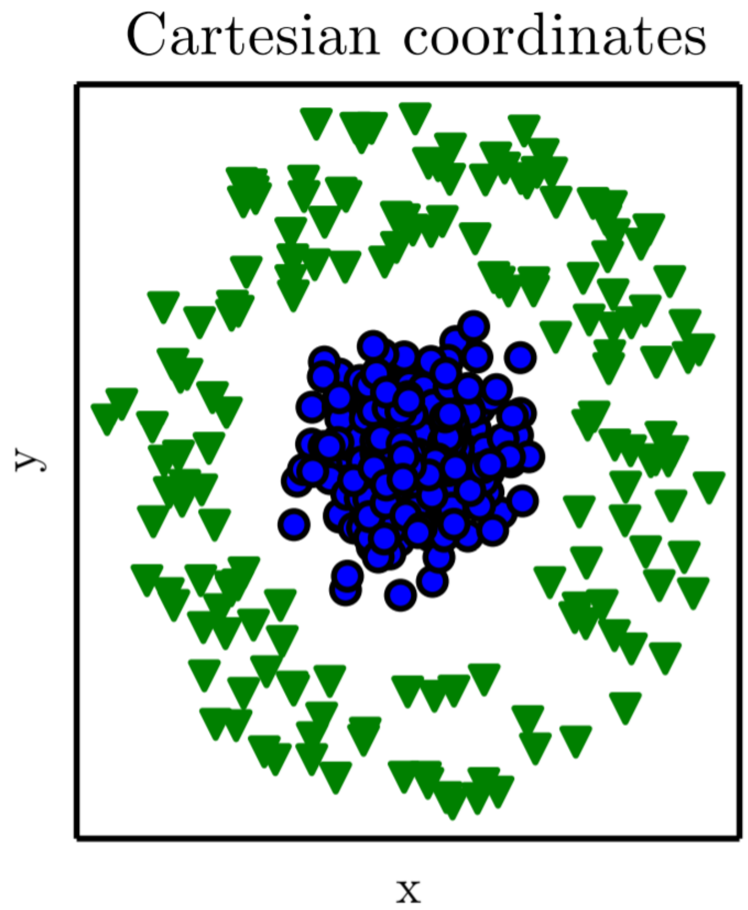


Source: Amini and Soleimany, MIT 6.S191: Introduction to Deep Learning

Representation matters

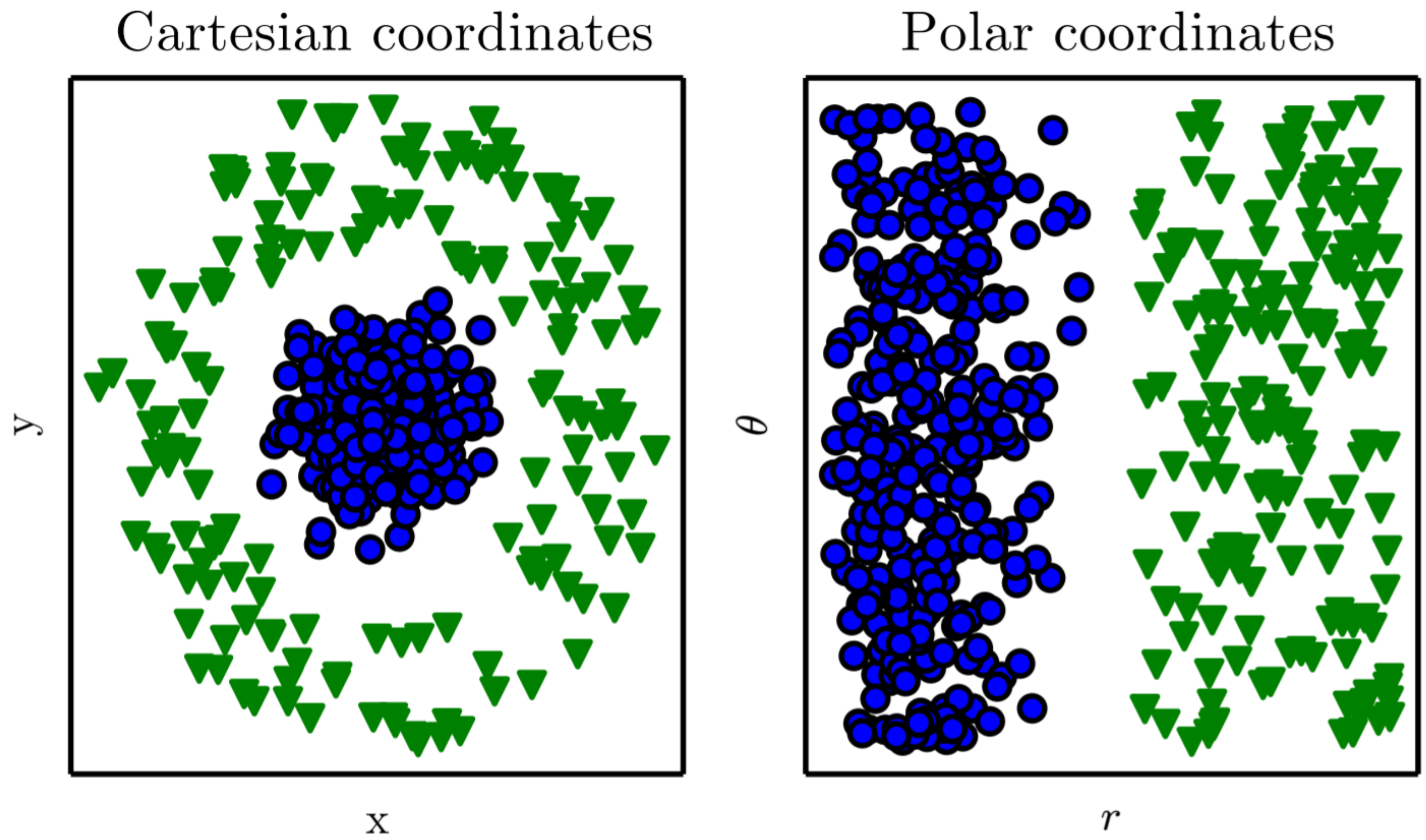
Source: Ian Goodfellow, Yoshua Bengio and Aaron Courville, Deep Learning, MIT Press

Representation matters



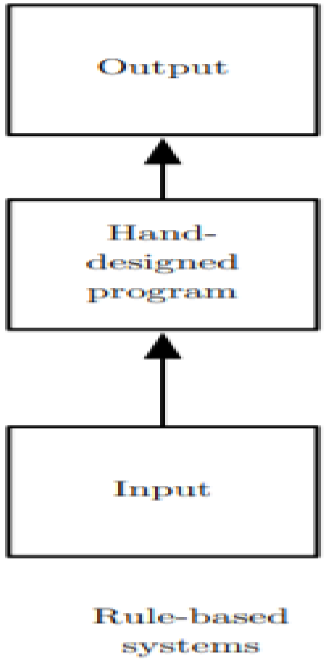
Source: Ian Goodfellow, Yoshua Bengio and Aaron Courville, Deep Learning, MIT Press

Representation matters



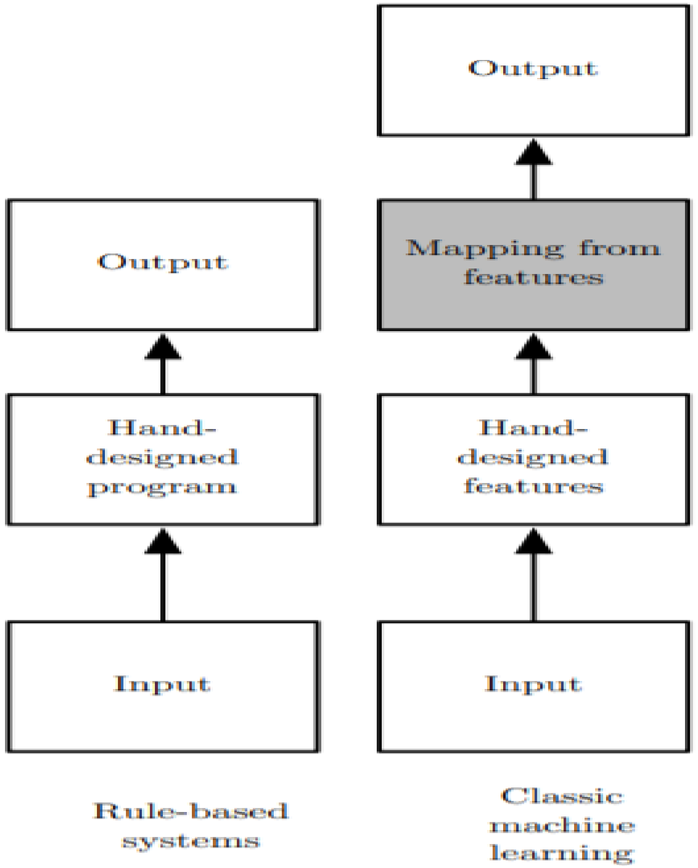
Source: Ian Goodfellow, Yoshua Bengio and Aaron Courville, Deep Learning, MIT Press

Learning representations from data



Source: Ian Goodfellow , Yoshua Bengio and Aaron Courville, Deep Learning, MIT Press

Learning representations from data



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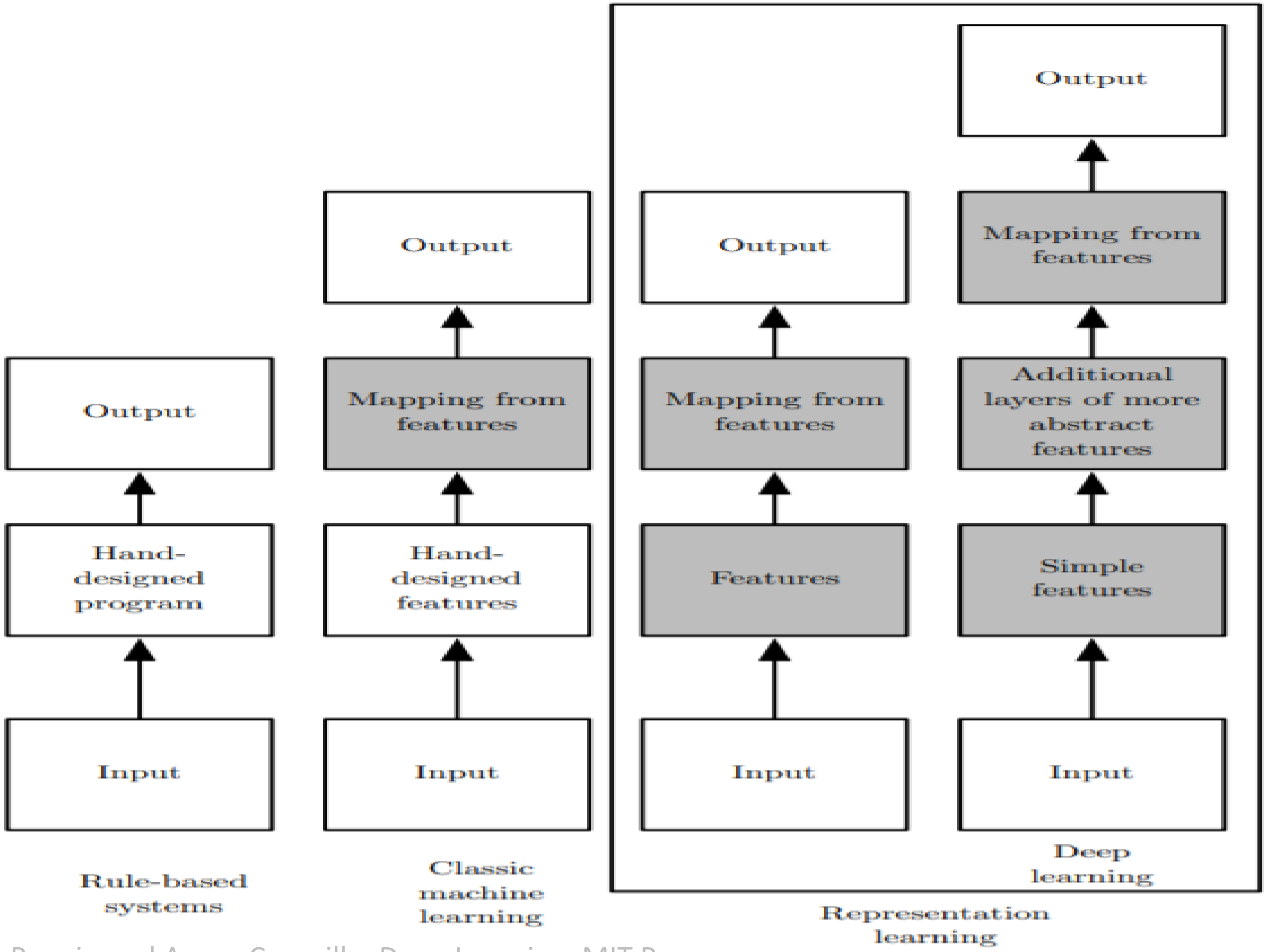
Learning representations from data



Source: Ian Goodfellow , Yoshua Bengio and Aaron Courville, Deep Learning, MIT Press

Depth: Repeated Composition

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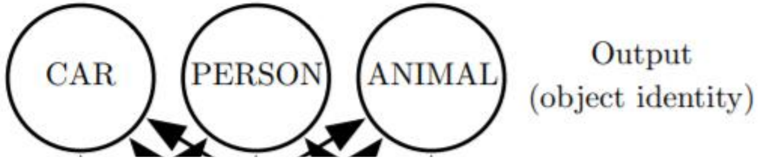

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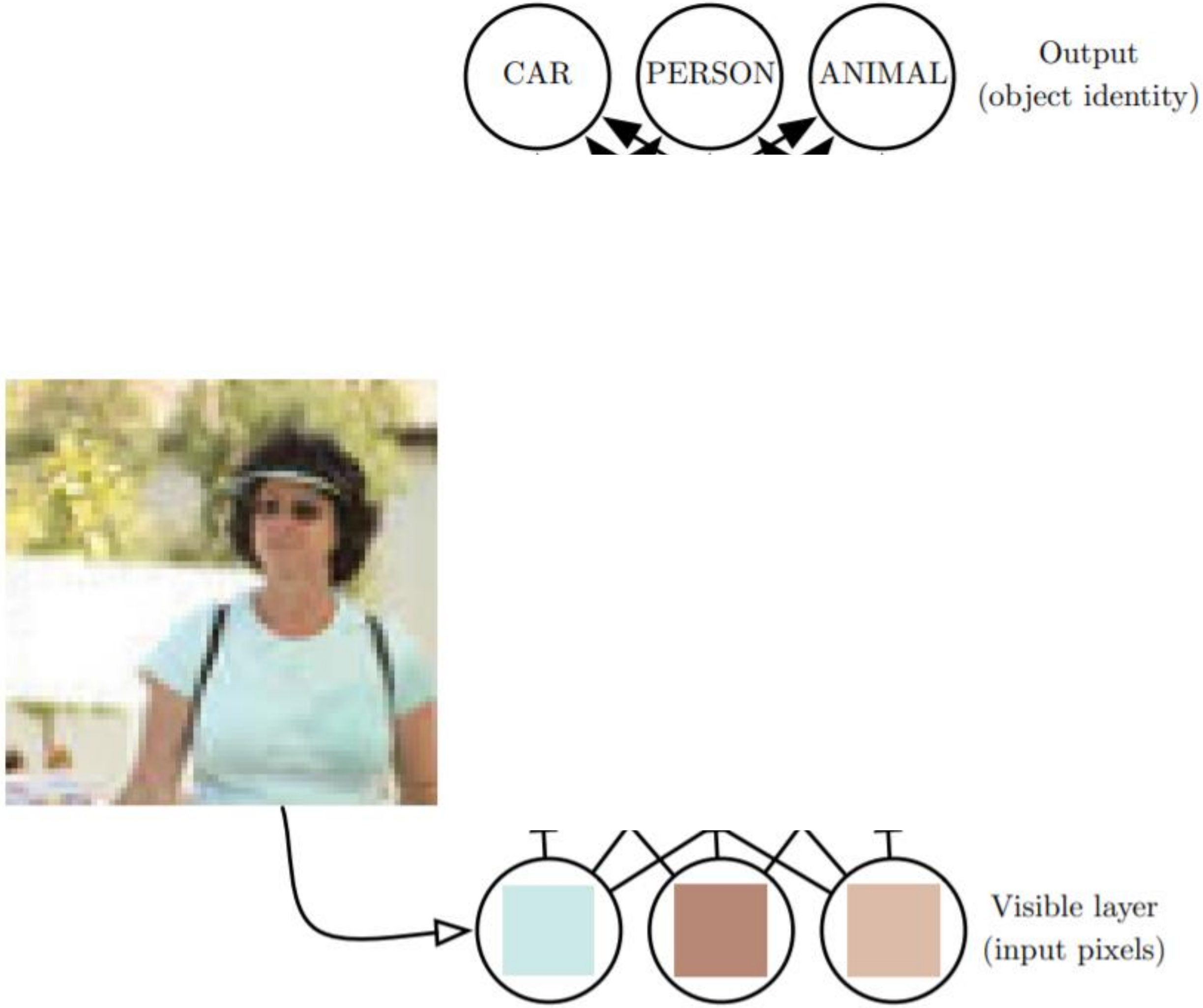

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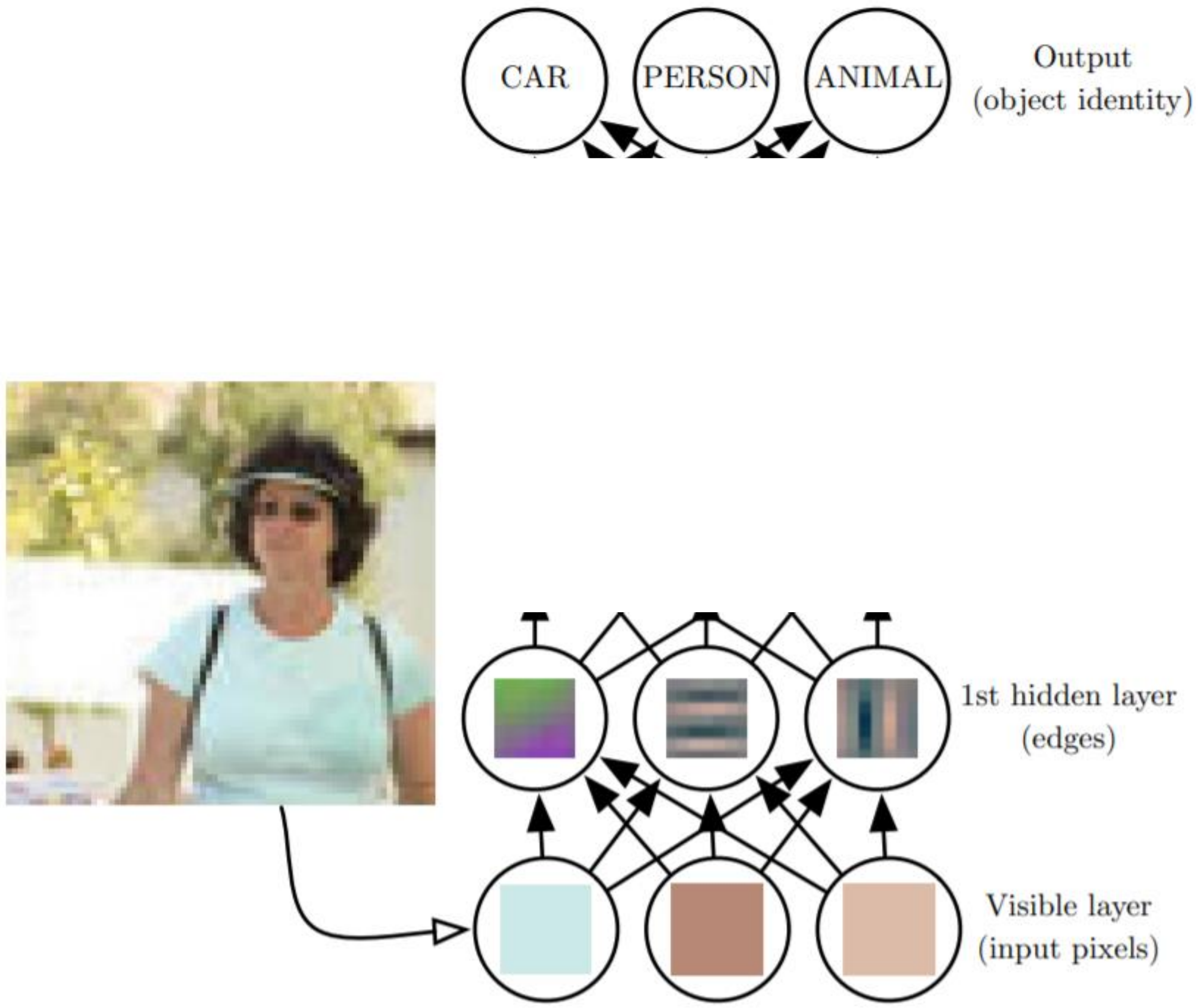

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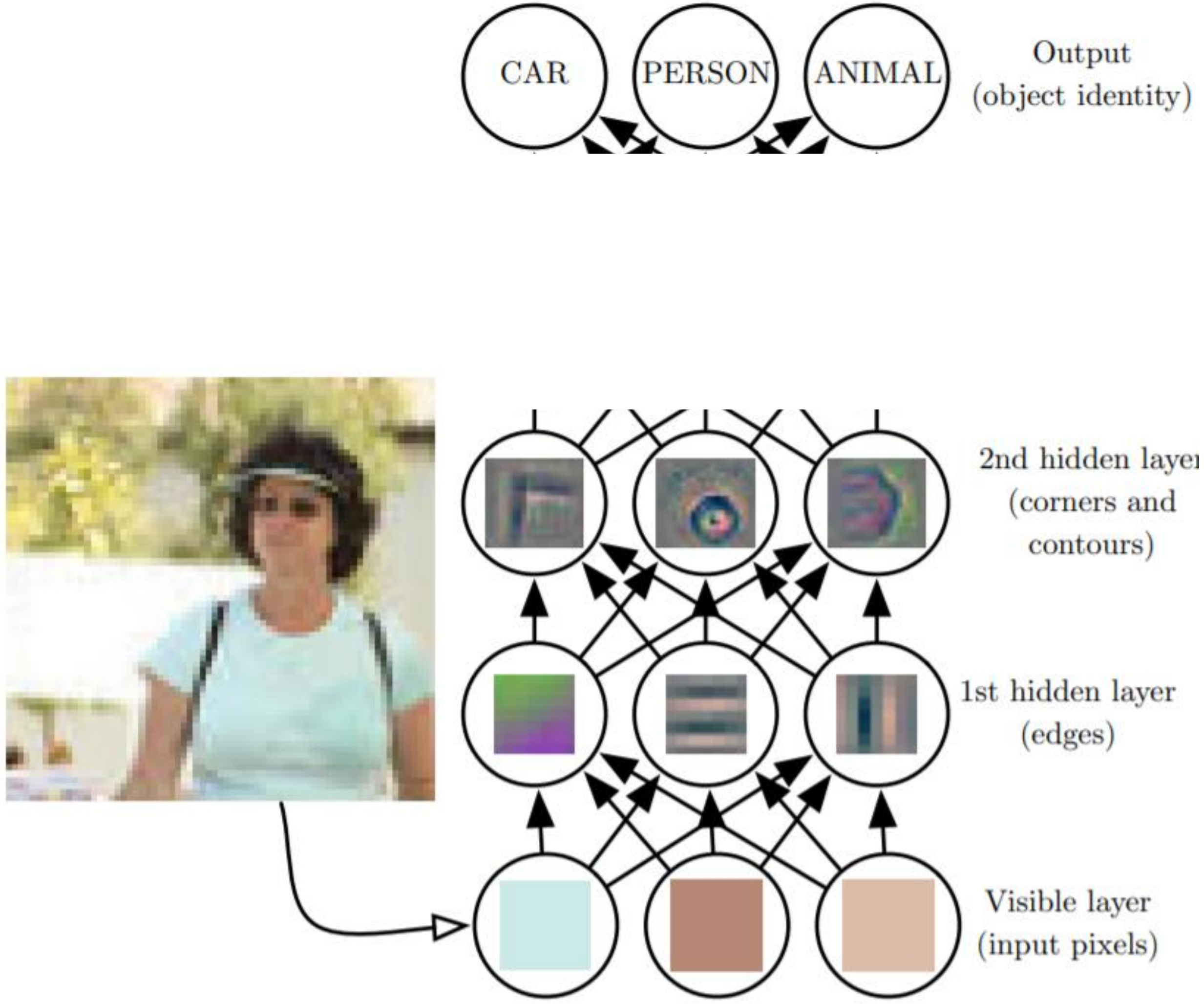

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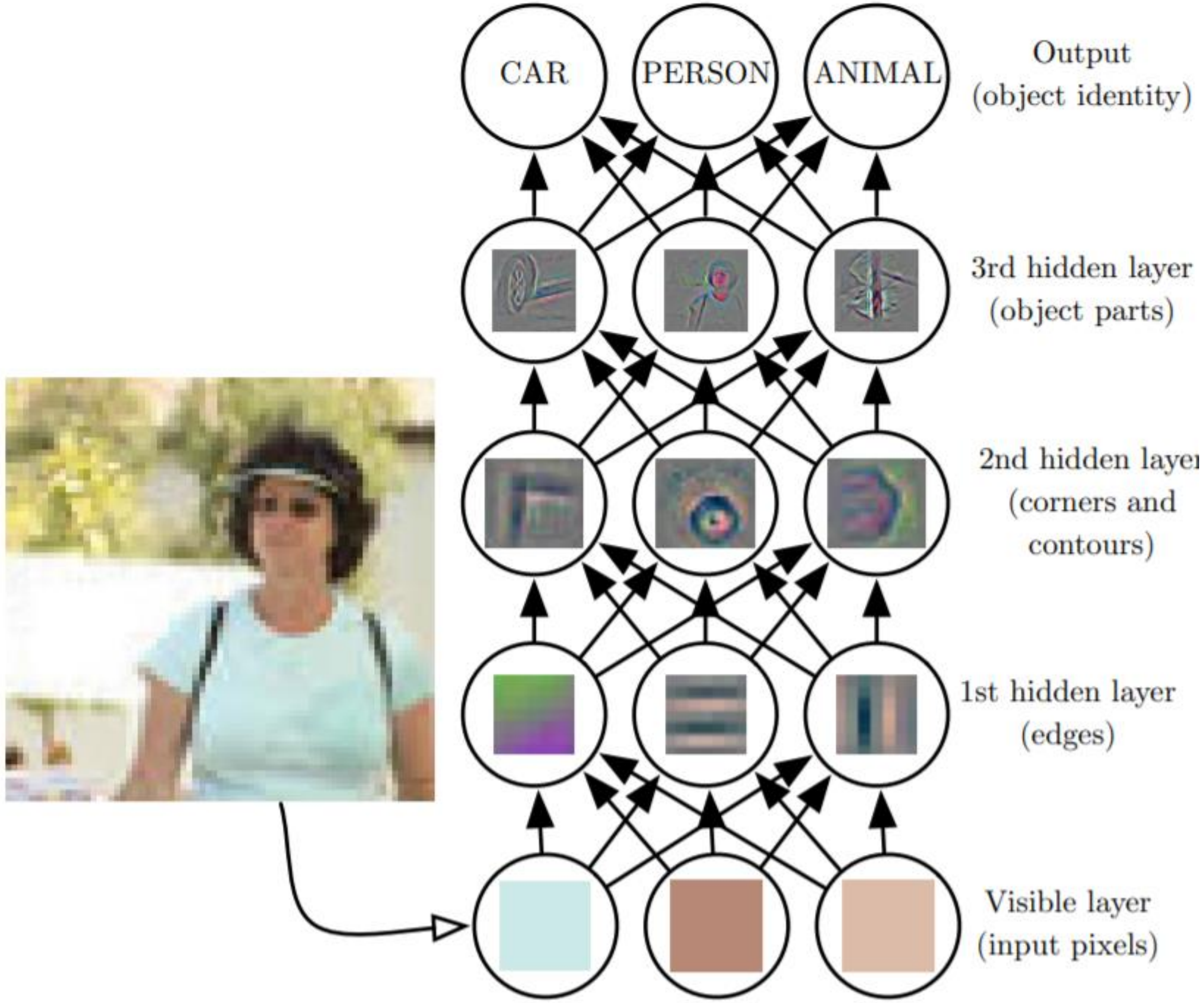

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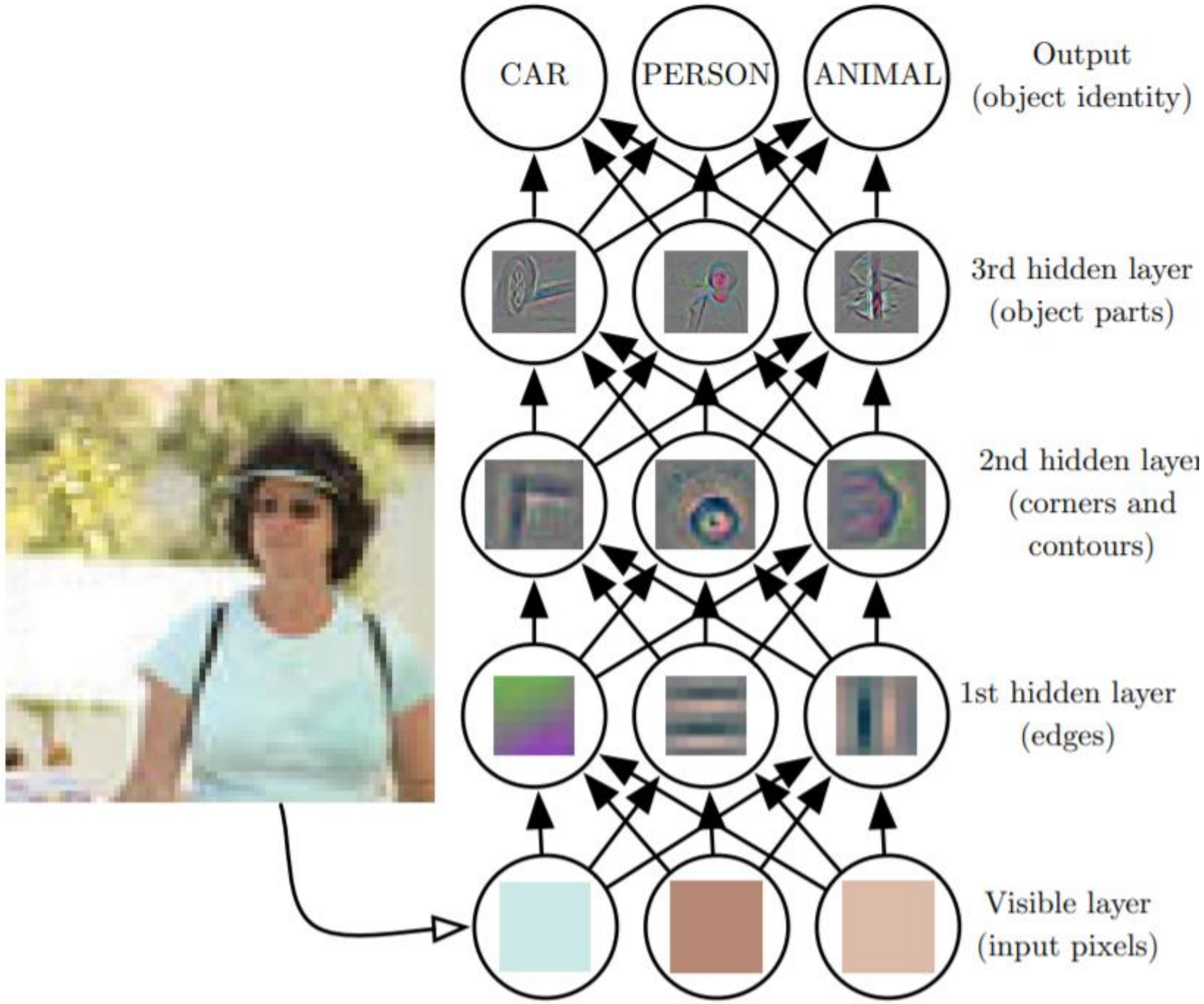

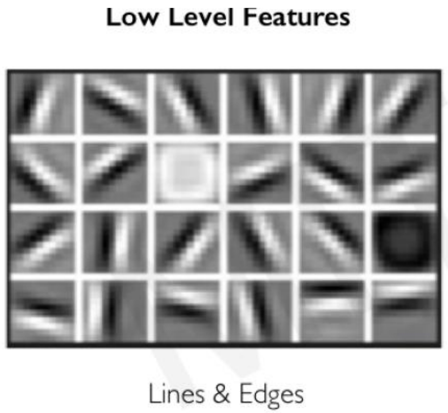

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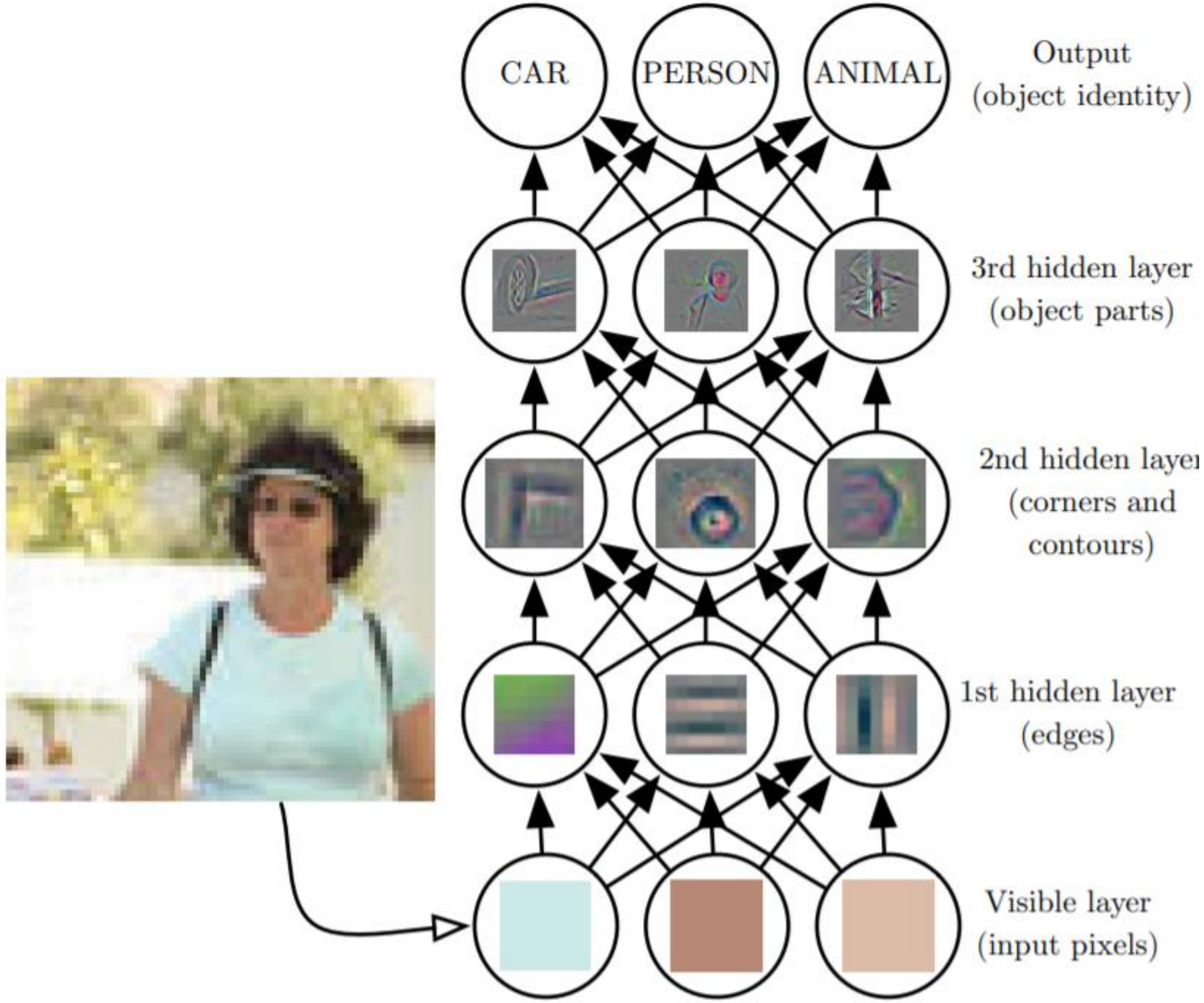

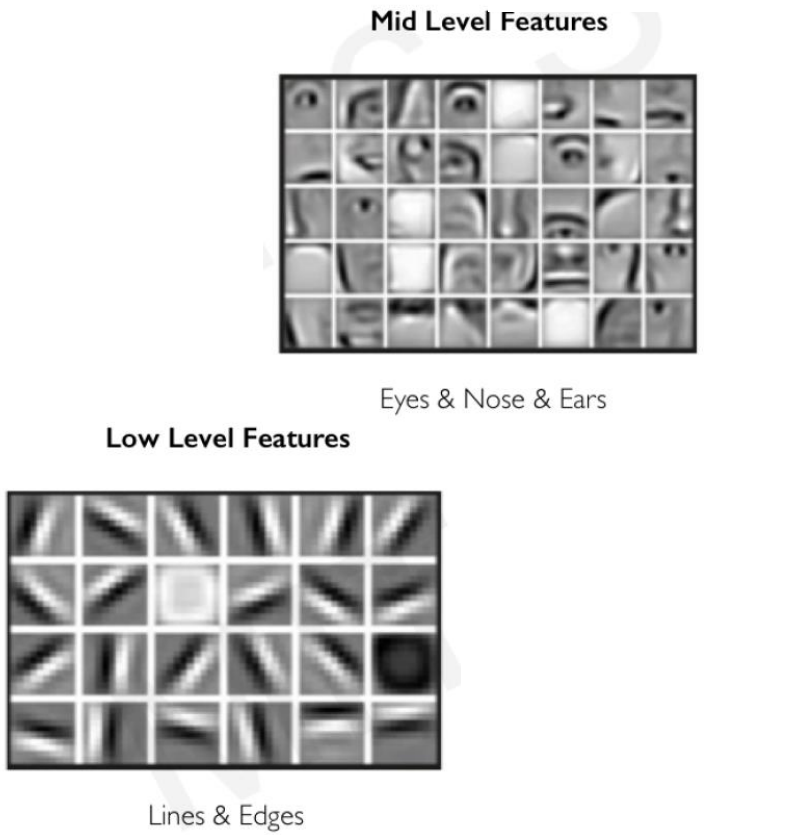

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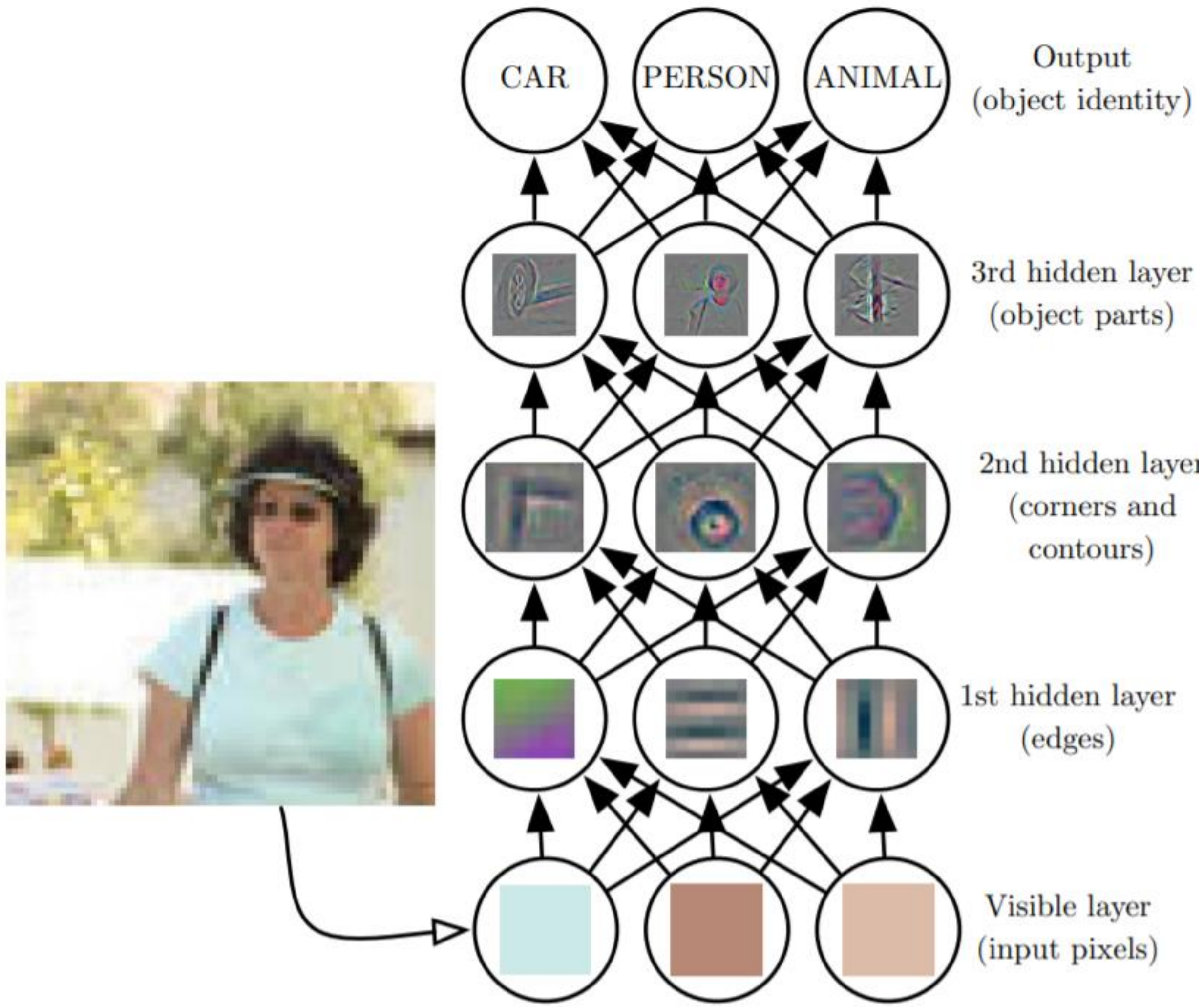

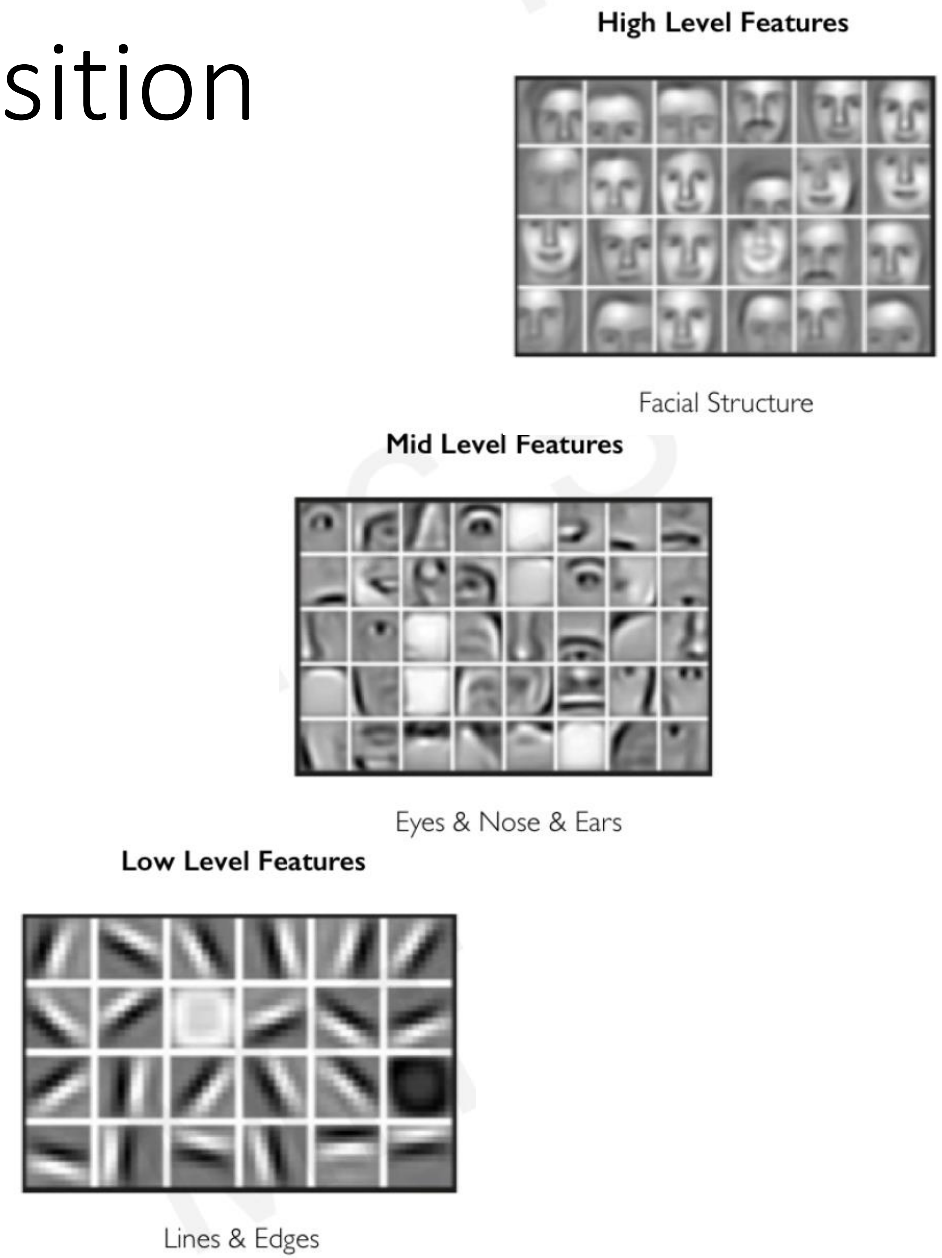

Source: Ian Goodfellow , Yoshua Bengio and Aaron Courville, Deep Learning, MIT Press
Adapted from: Amini and Soleimany, MIT 6.S191: Introduction to Deep Learning

Image Classification

Input image

Source: Niessner and Leal-Taixé, Introduction to Deep Learning

Image Classification

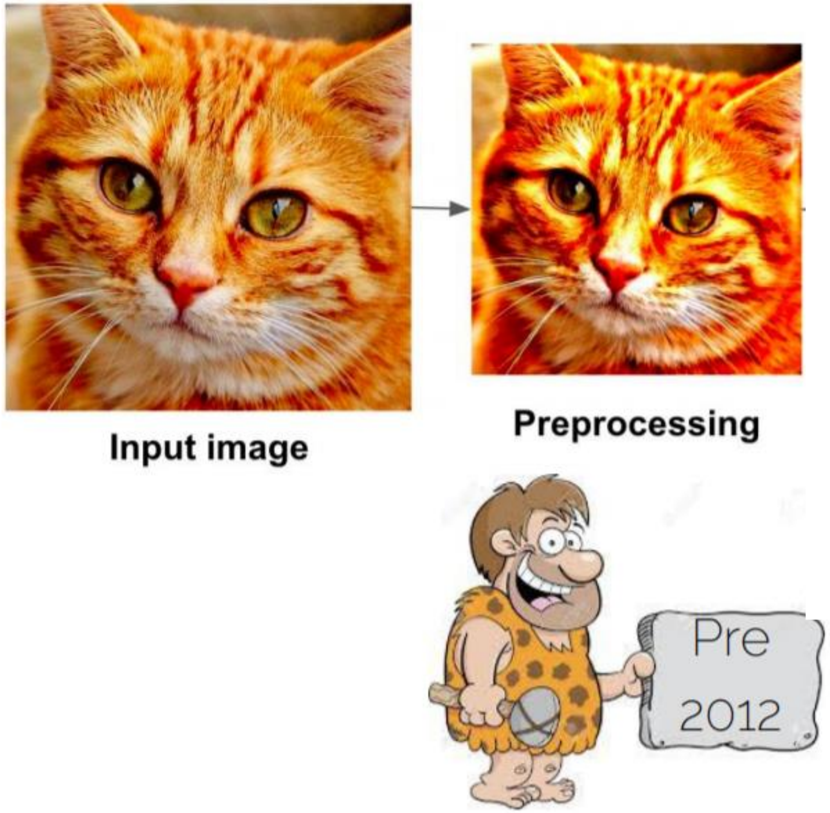

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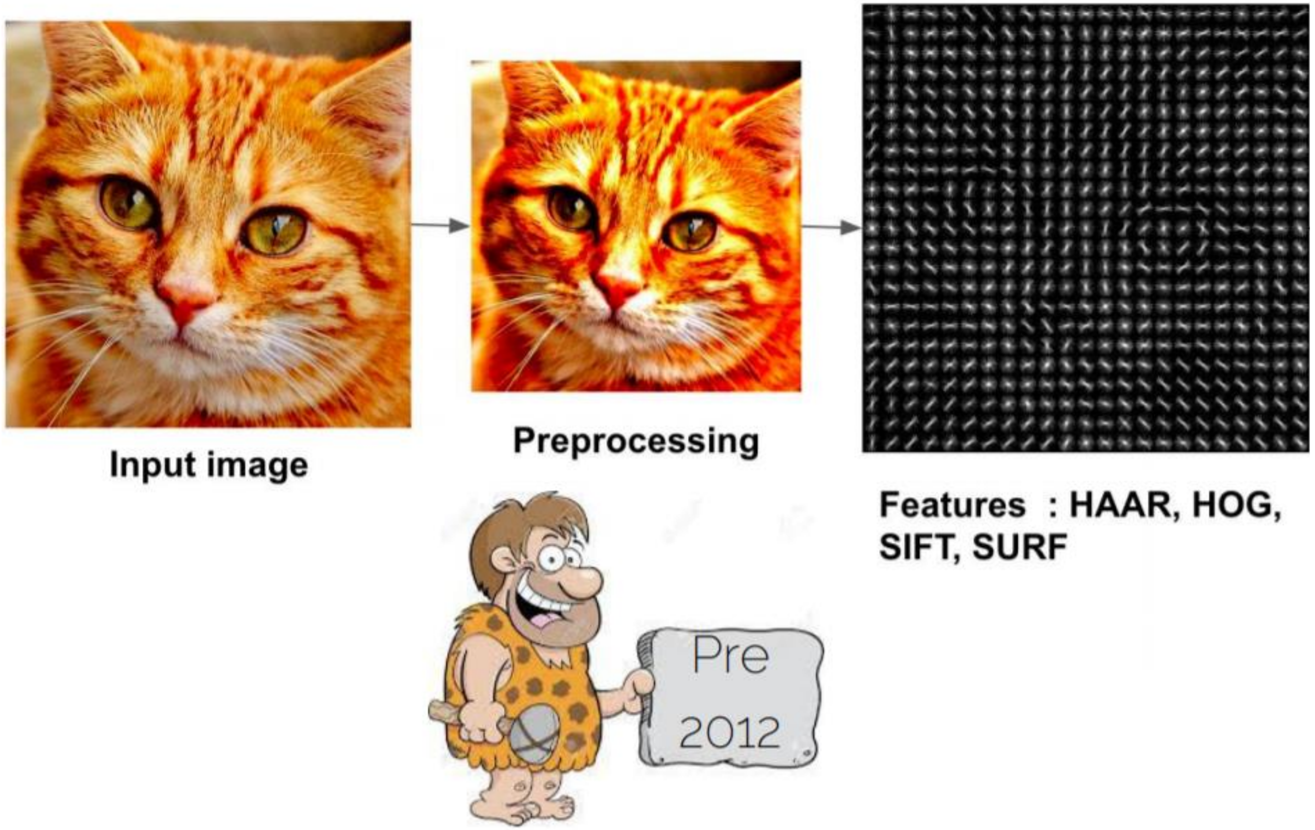
Image Classification



Source: Niessner and Leal-Taixé, Introduction to Deep Learning

Image Classification

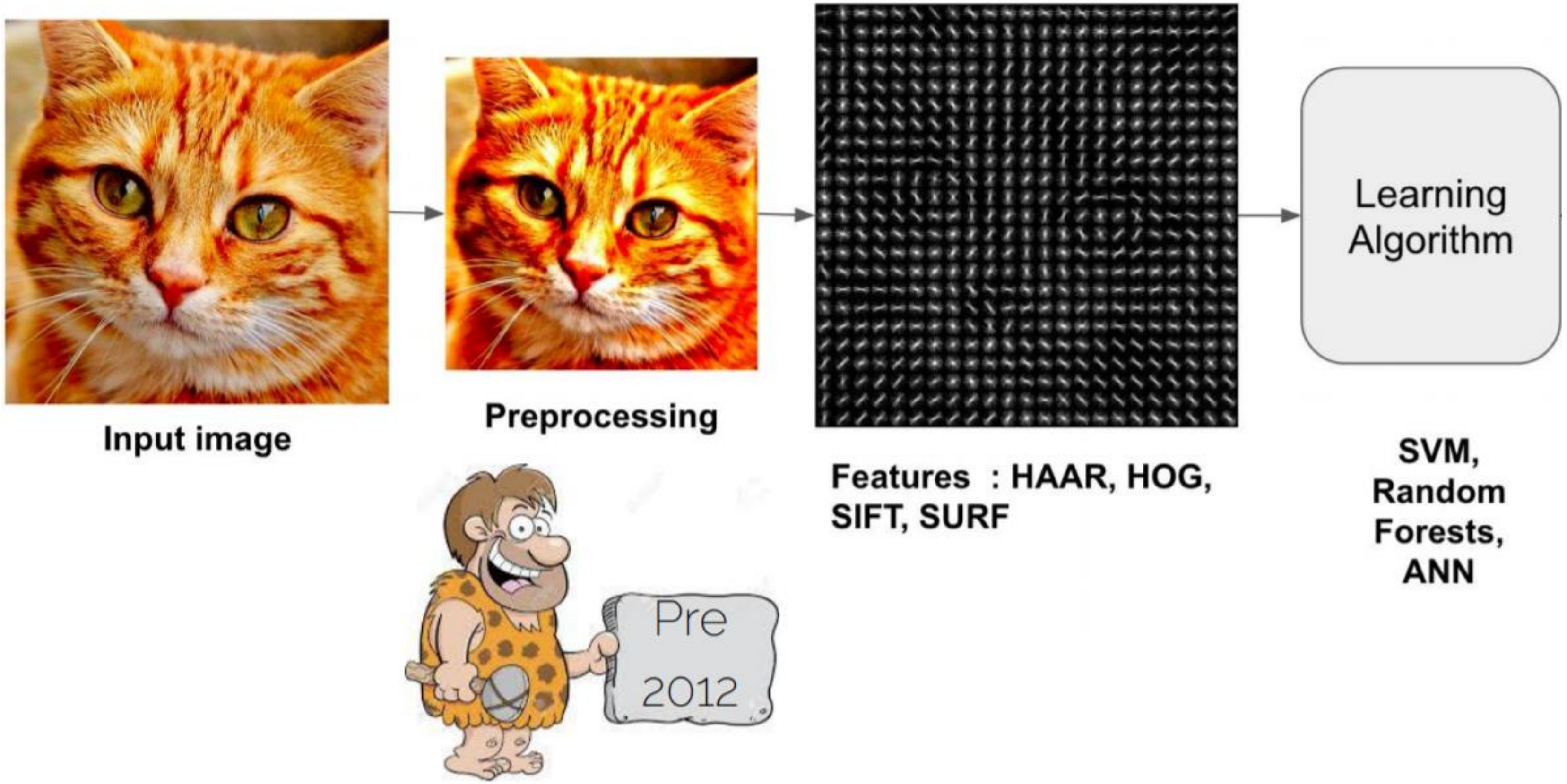


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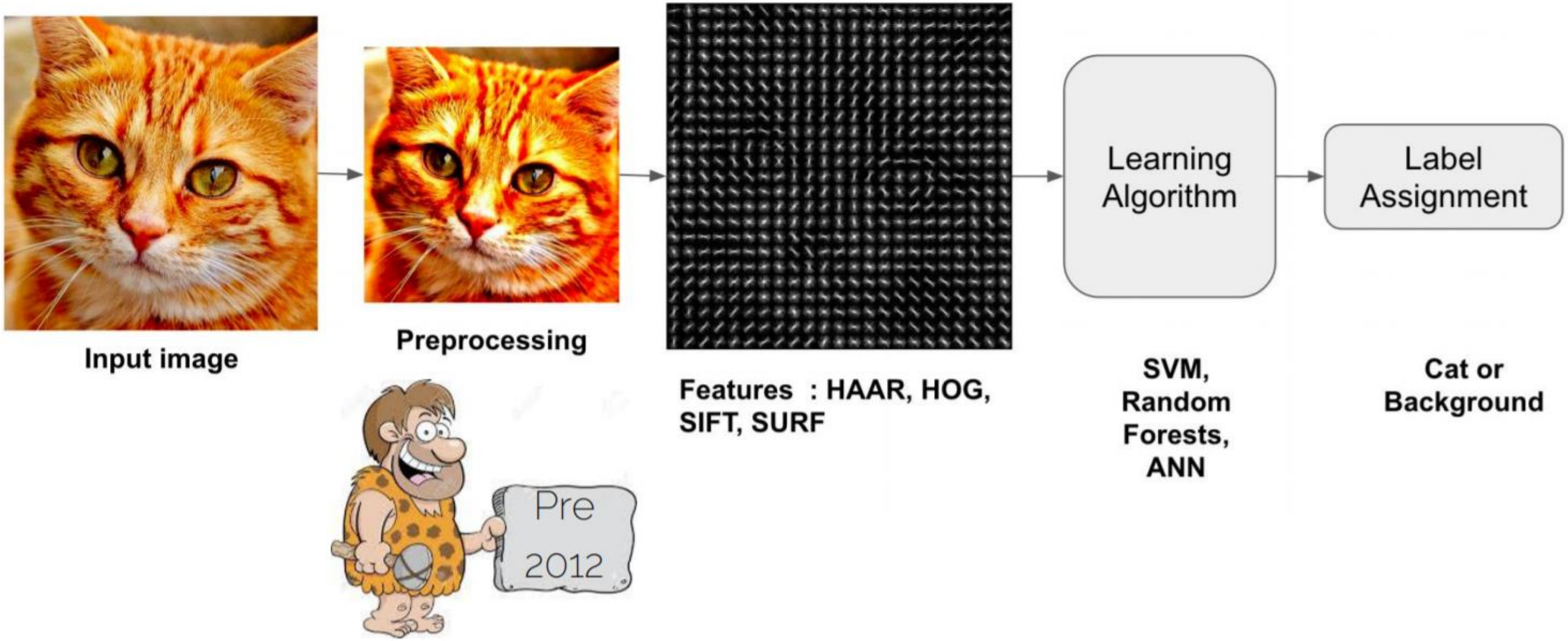
Image Classification



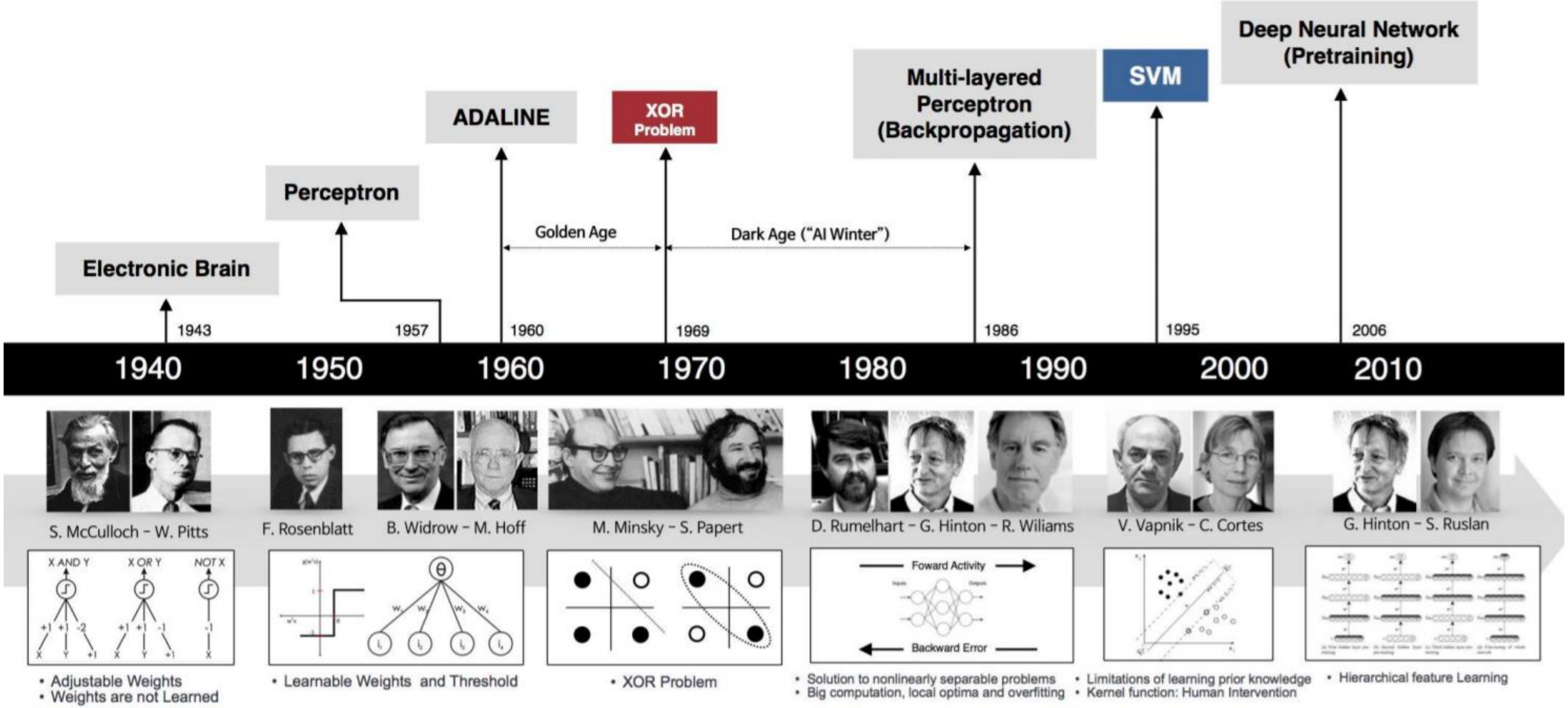
Source: Niessner and Leal-Taixé, Introduction to Deep Learning

Image Classification

Post 2012

Adapted from: Niessner and Leal-Taixé, Introduction to Deep Learning

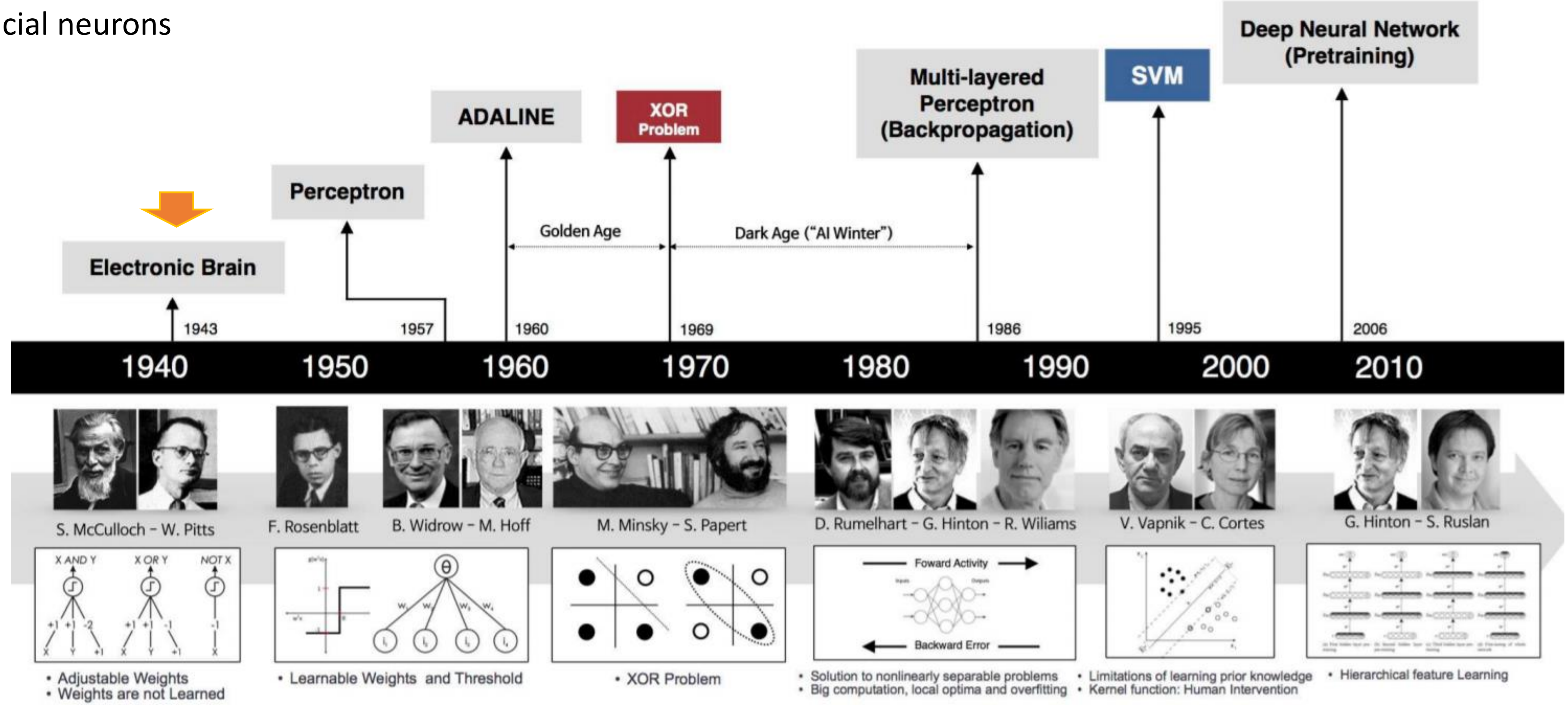
Deep Learning History



Source: Niessner and Leal-Taixé, Introduction to Deep Learning

Deep Learning History

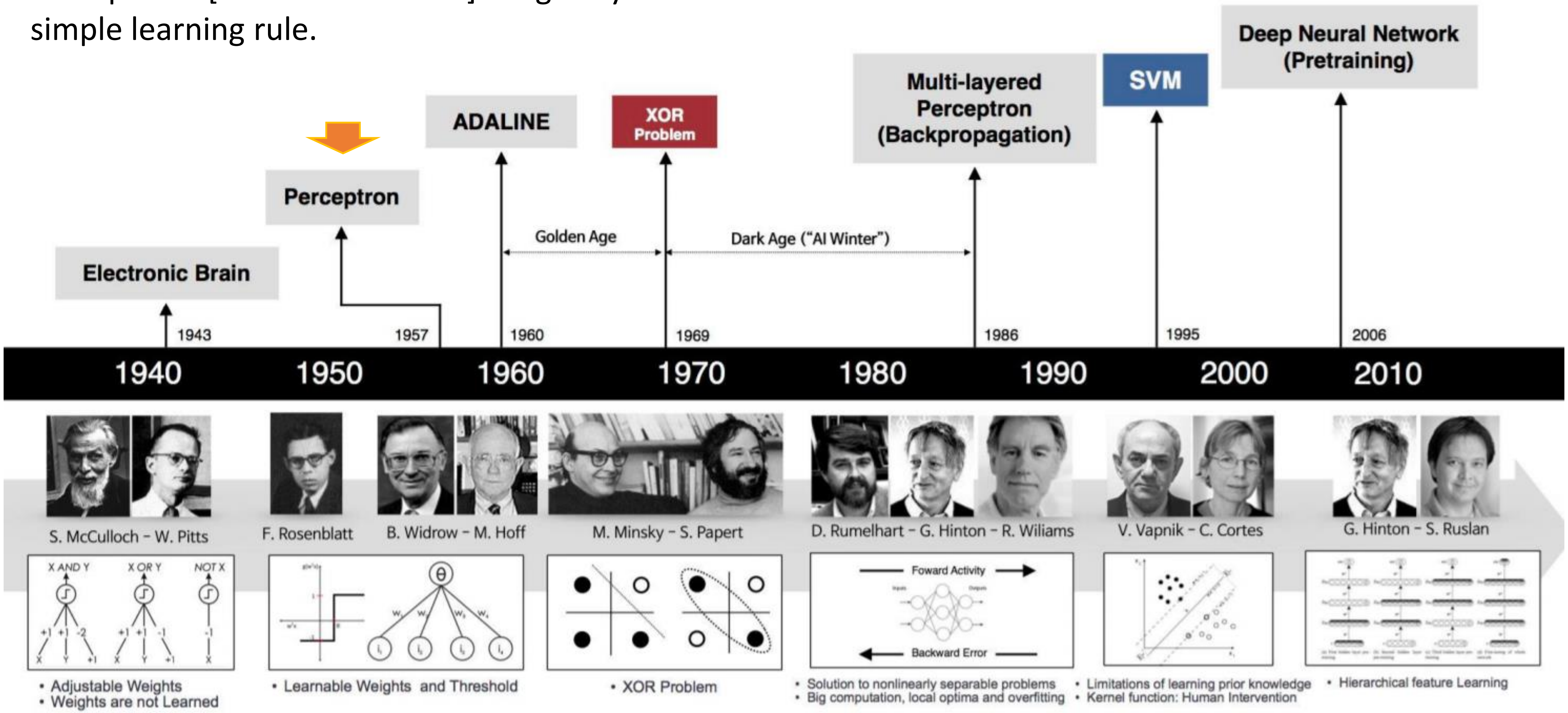
Early work in 1940's and 1950's by Hebb, McCulloch and Pitts on artificial neurons



Source: Niessner and Leal-Taixé, Introduction to Deep Learning

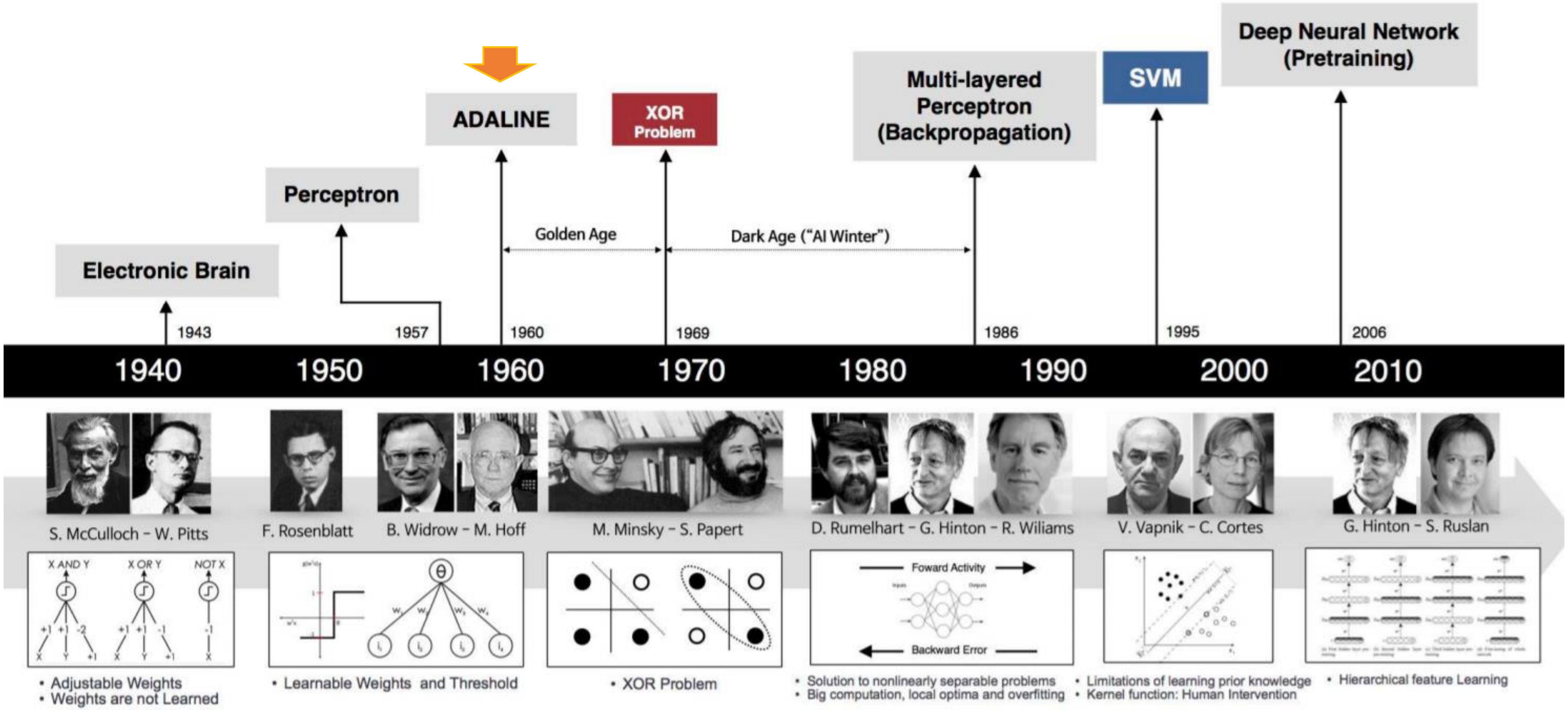
Deep Learning History

Perceptrons [Rosenblatt 1950's]. Single layer networks with simple learning rule.



Source: Niessner and Leal-Taixé, Introduction to Deep Learning

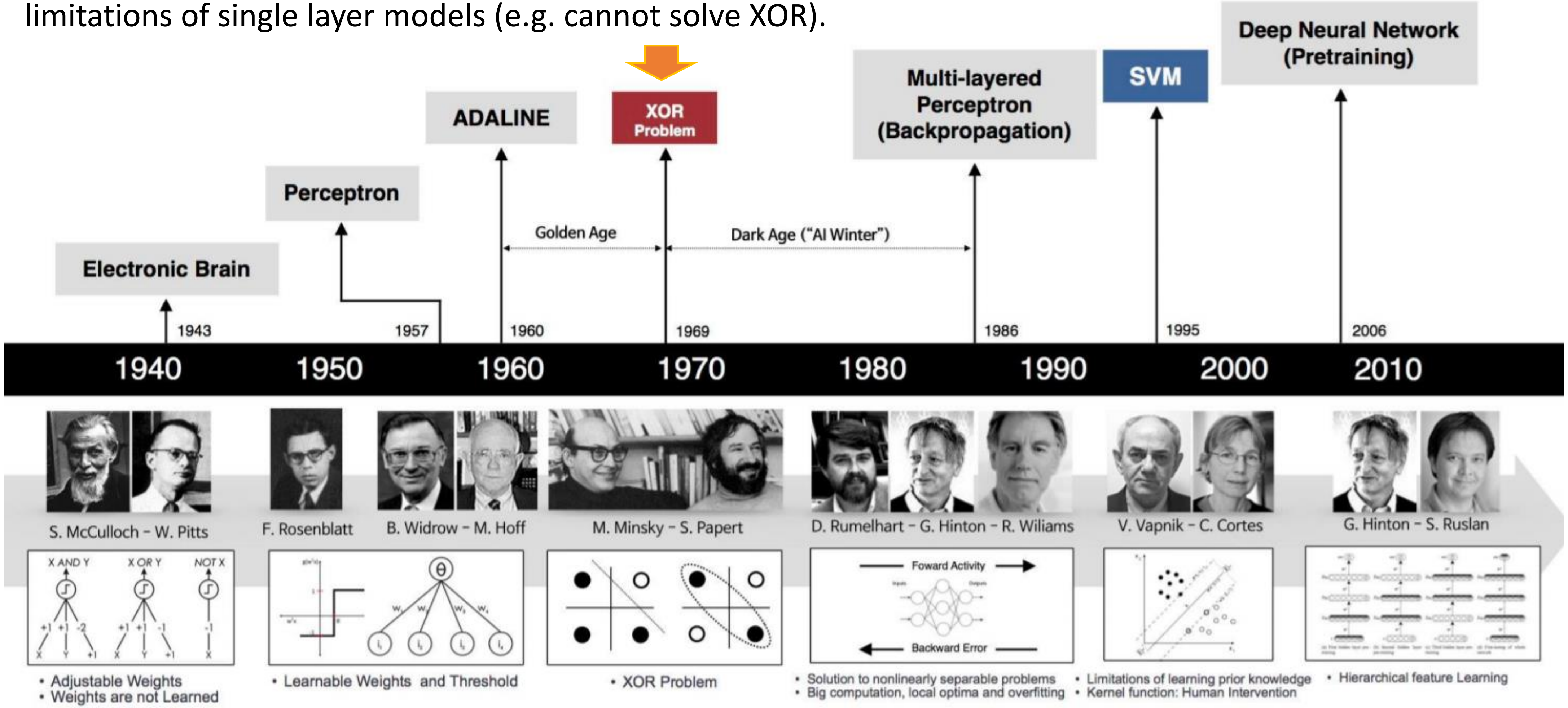
Deep Learning History



Source: Niessner and Leal-Taixé, Introduction to Deep Learning

Deep Learning History

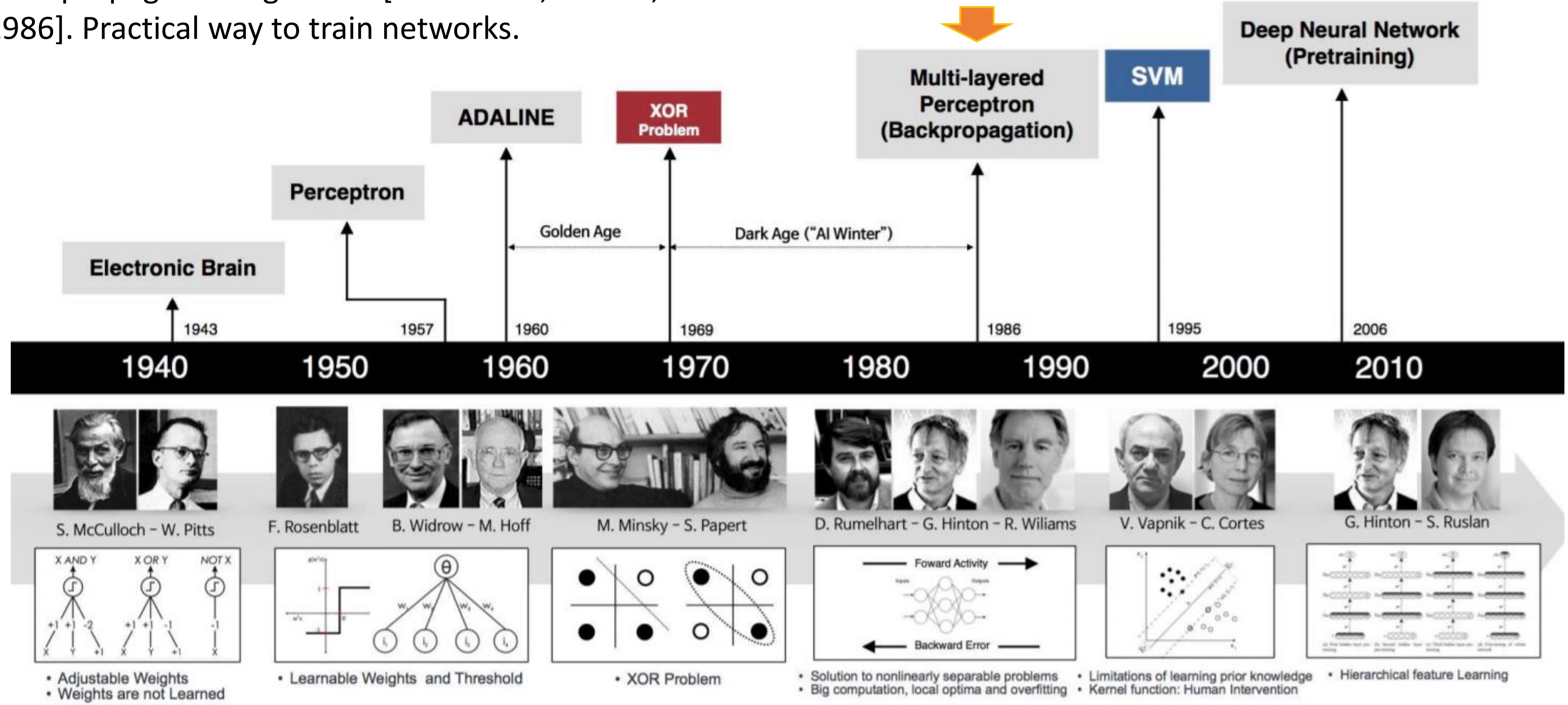
Perceptron book [Minsky and Pappert 1969]. Showed limitations of single layer models (e.g. cannot solve XOR).



Source: Niessner and Leal-Taixé, Introduction to Deep Learning

Deep Learning History

Back-propagation algorithm [Rumelhart, Hinton, Williams 1986]. Practical way to train networks.



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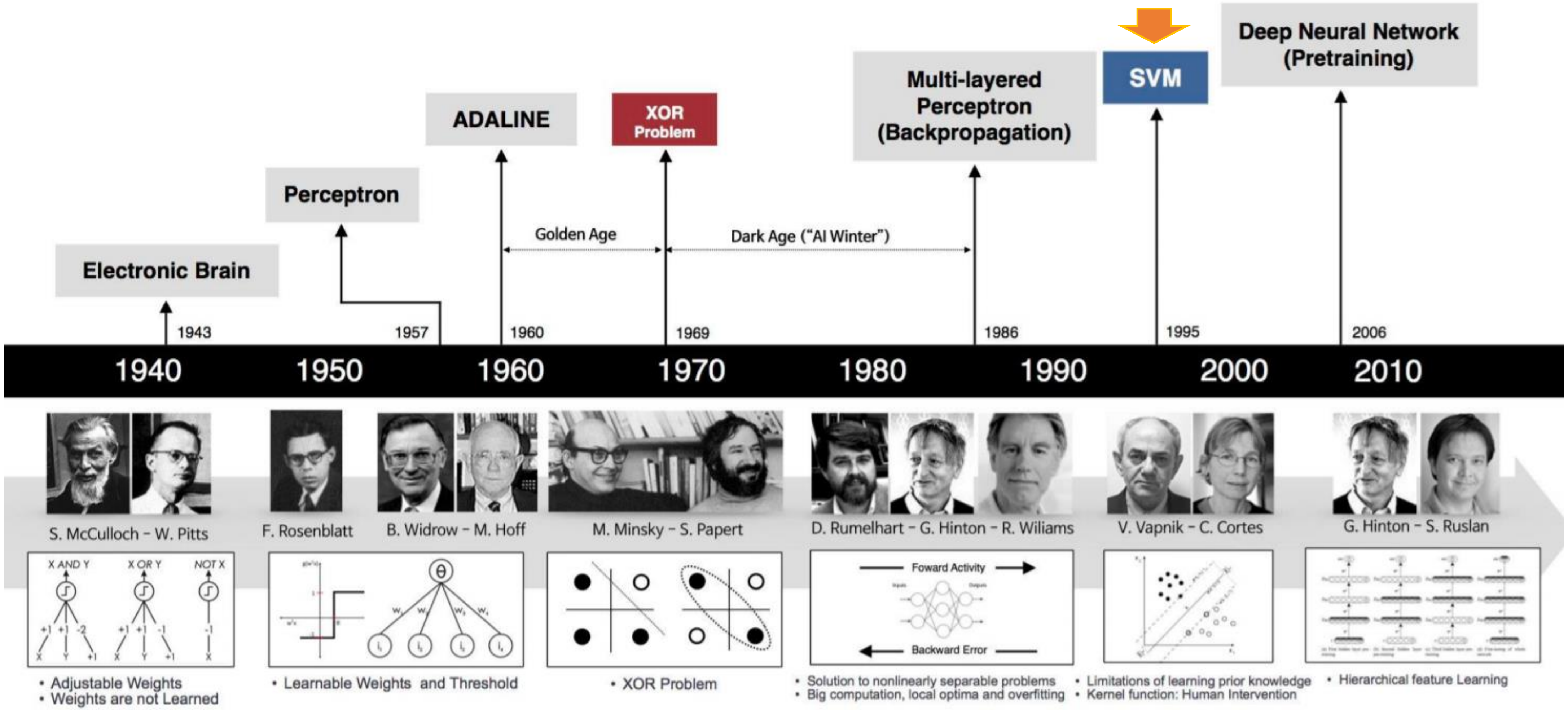
Deep Learning History



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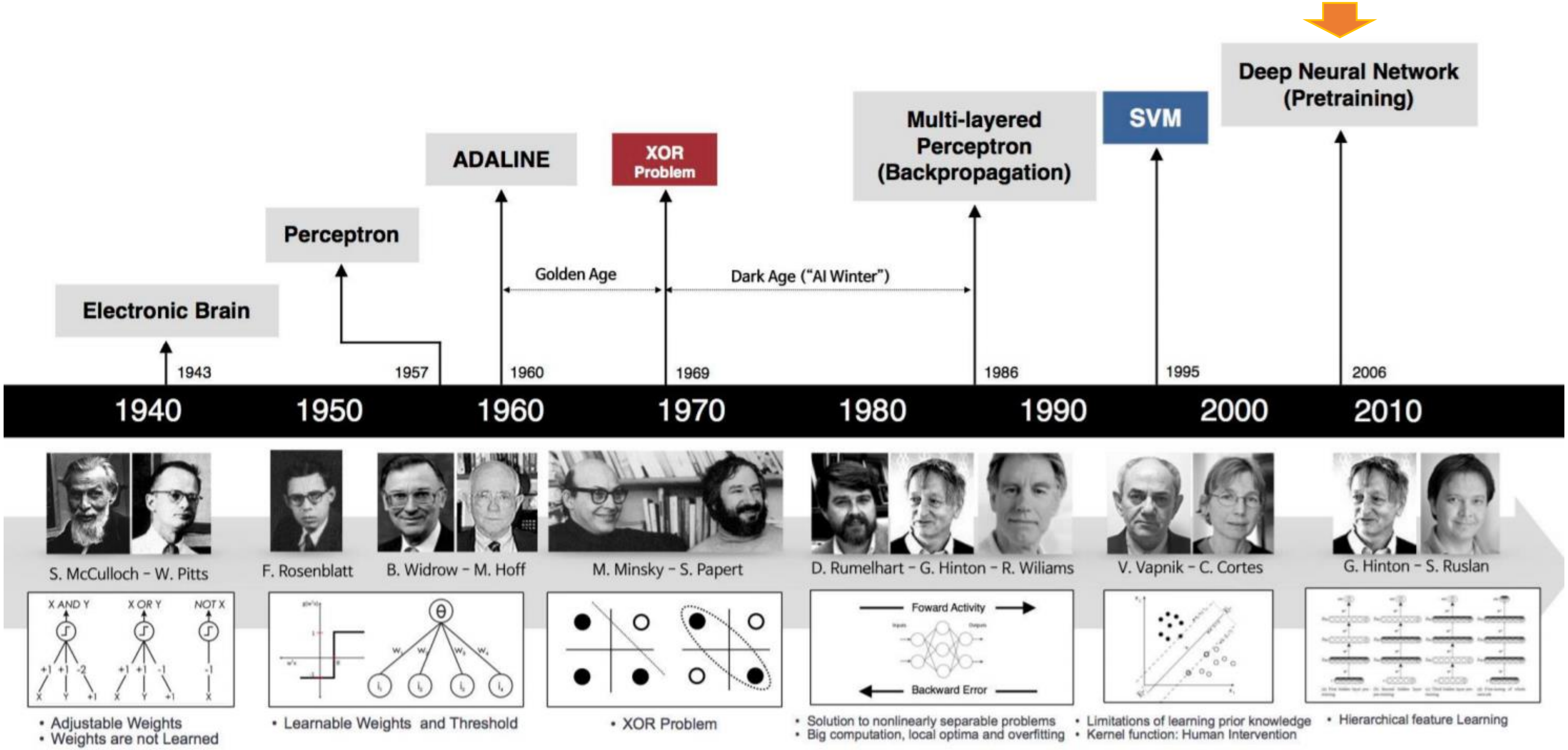
Deep Learning History

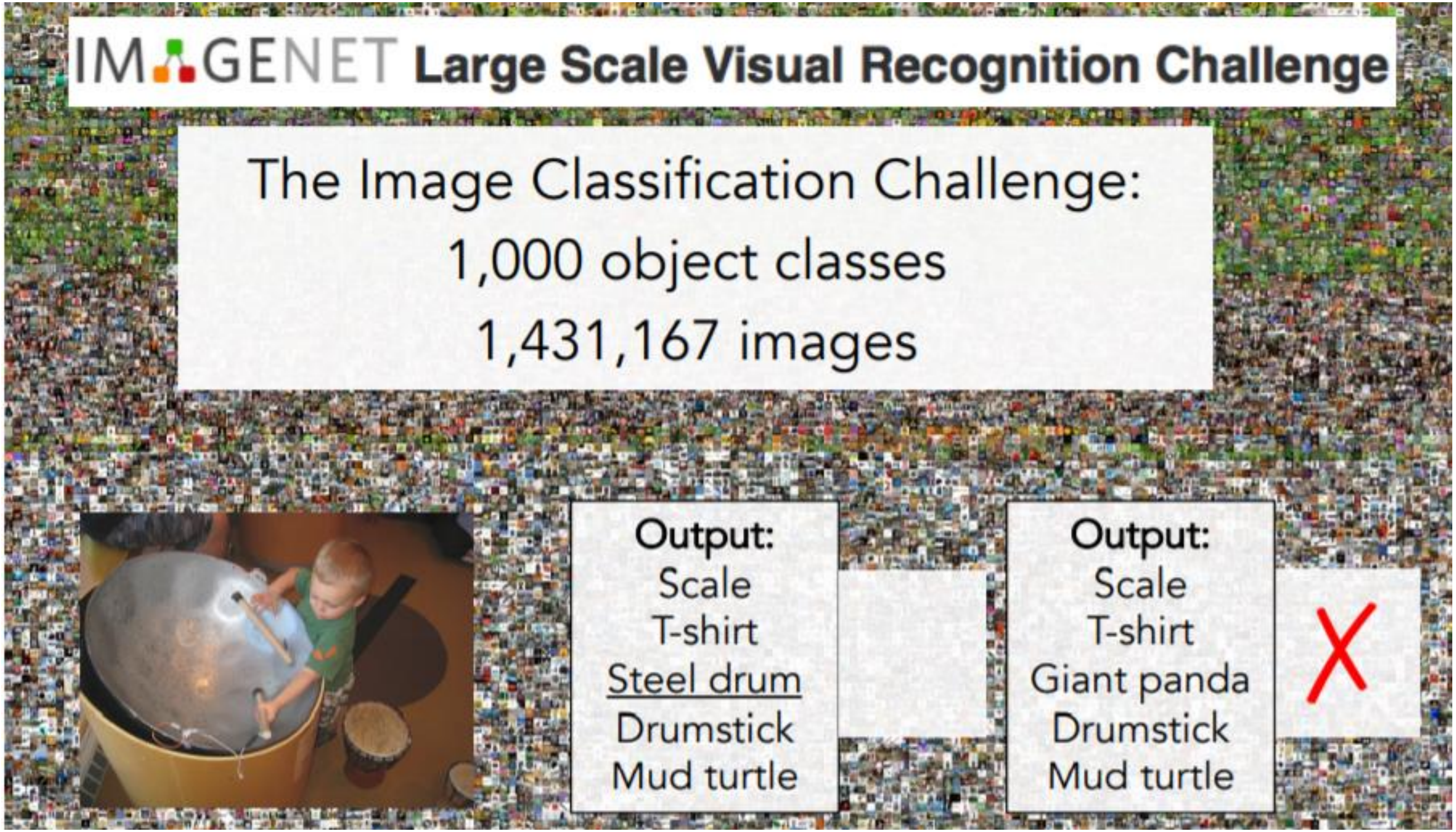


Source: Niessner and Leal-Taixé, Introduction to Deep Learning

Application: ImageNet

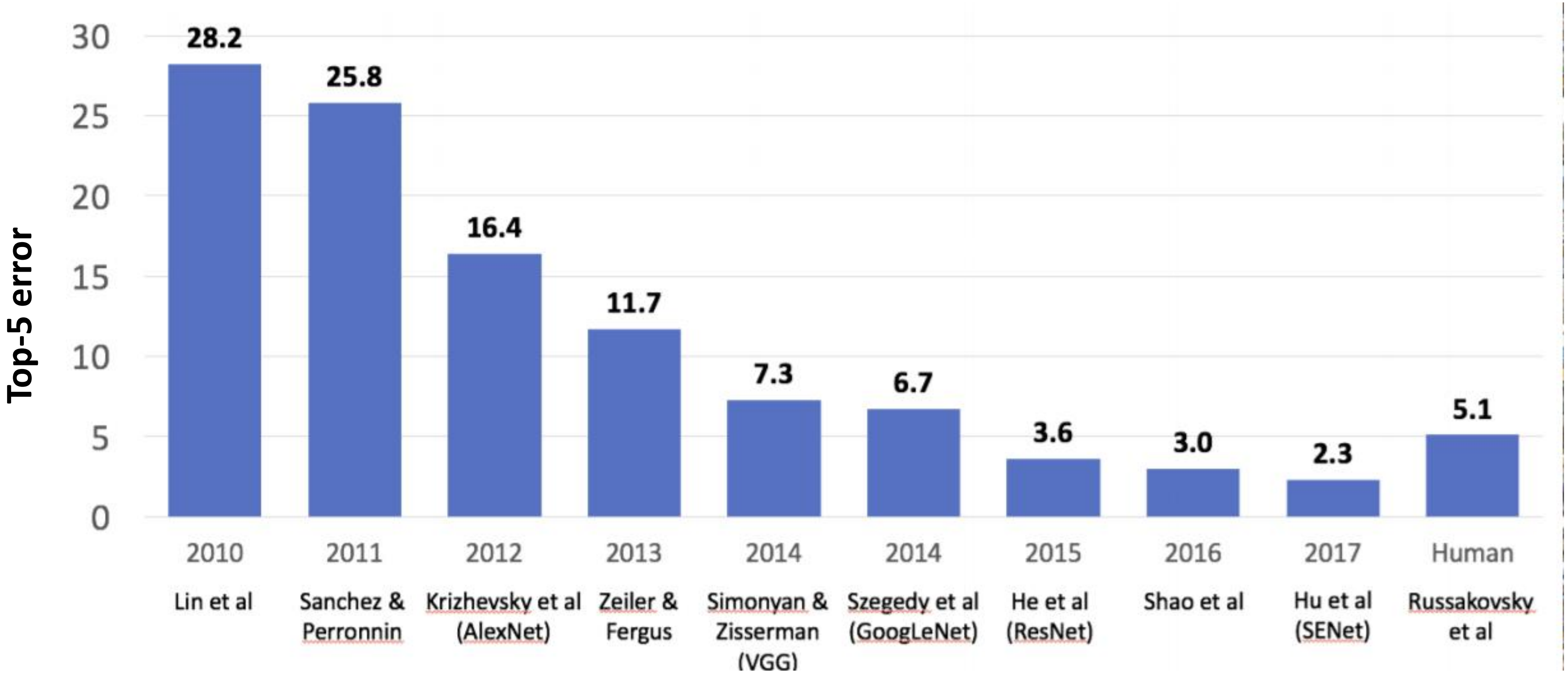
[Deng et al. CVPR 2009]

- ~14 million labeled images, 20k classes
- Images gathered from Internet
- Human labels via Amazon Turk



Source: Fei-Fei Li, Ranjay Krishna, Danfei Xu, CS231n: Convolutional Neural Networks for Visual Recognition

AlexNet


Top-5 error
30
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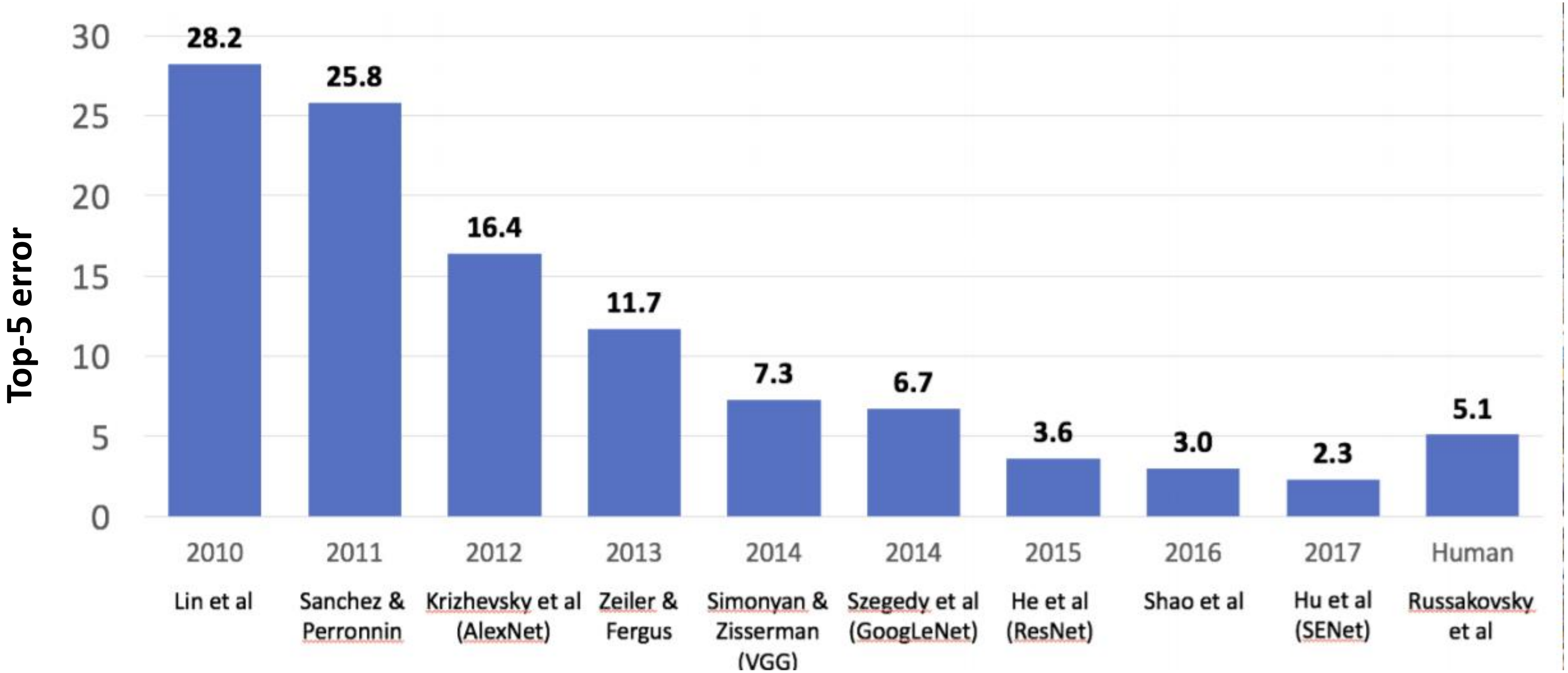
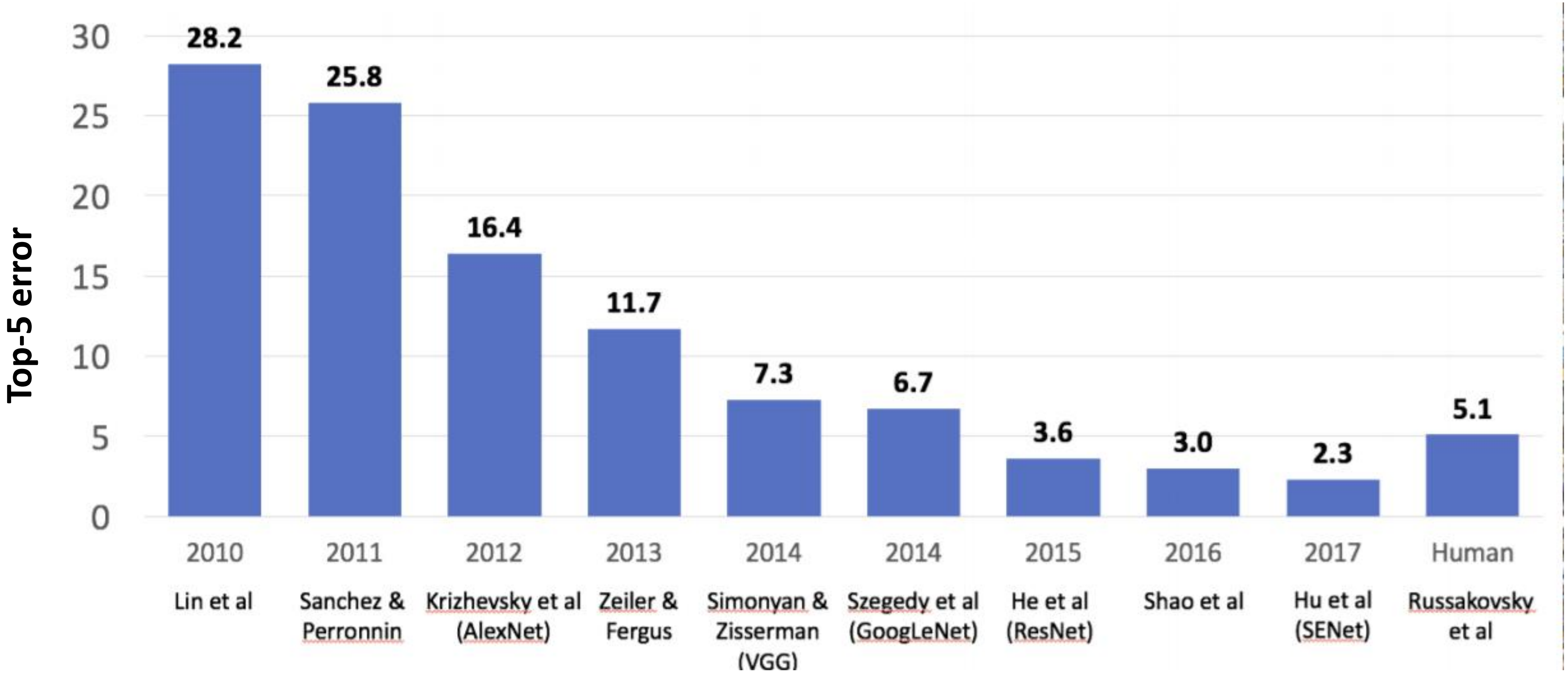
20
15
10
5
0
28.2
25.8
16.4
11.7
7.3
6.7
3.6
3.0
2.3
5.1
2010
Lin et al
2011
Sanchez & Perronnin
2012
Krizhevsky et al (AlexNet)
2013
Zeiler & Fergus
2014
Simonyan & Zisserman (VGG)
2014
Szegedy et al (GoogLeNet)
2015
He et al (ResNet)
2016
Shao et al
2017
Hu et al (SENet)
Human
Russakovsky et al

AlexNet



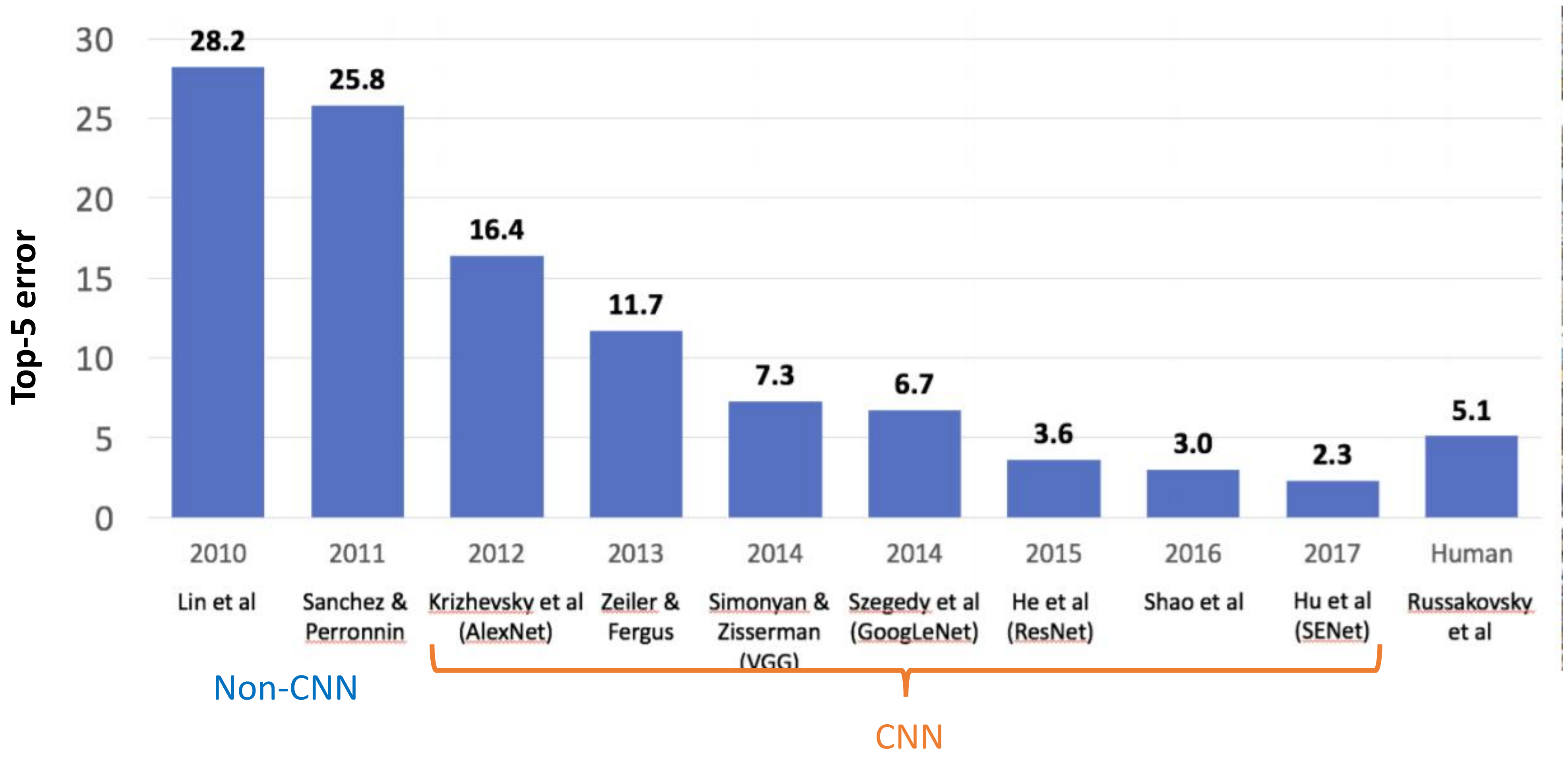
Top-5 error: percentage of test samples for which the correct class was not in the top 5 predicted classes

AlexNet



Non-CNN

AlexNet



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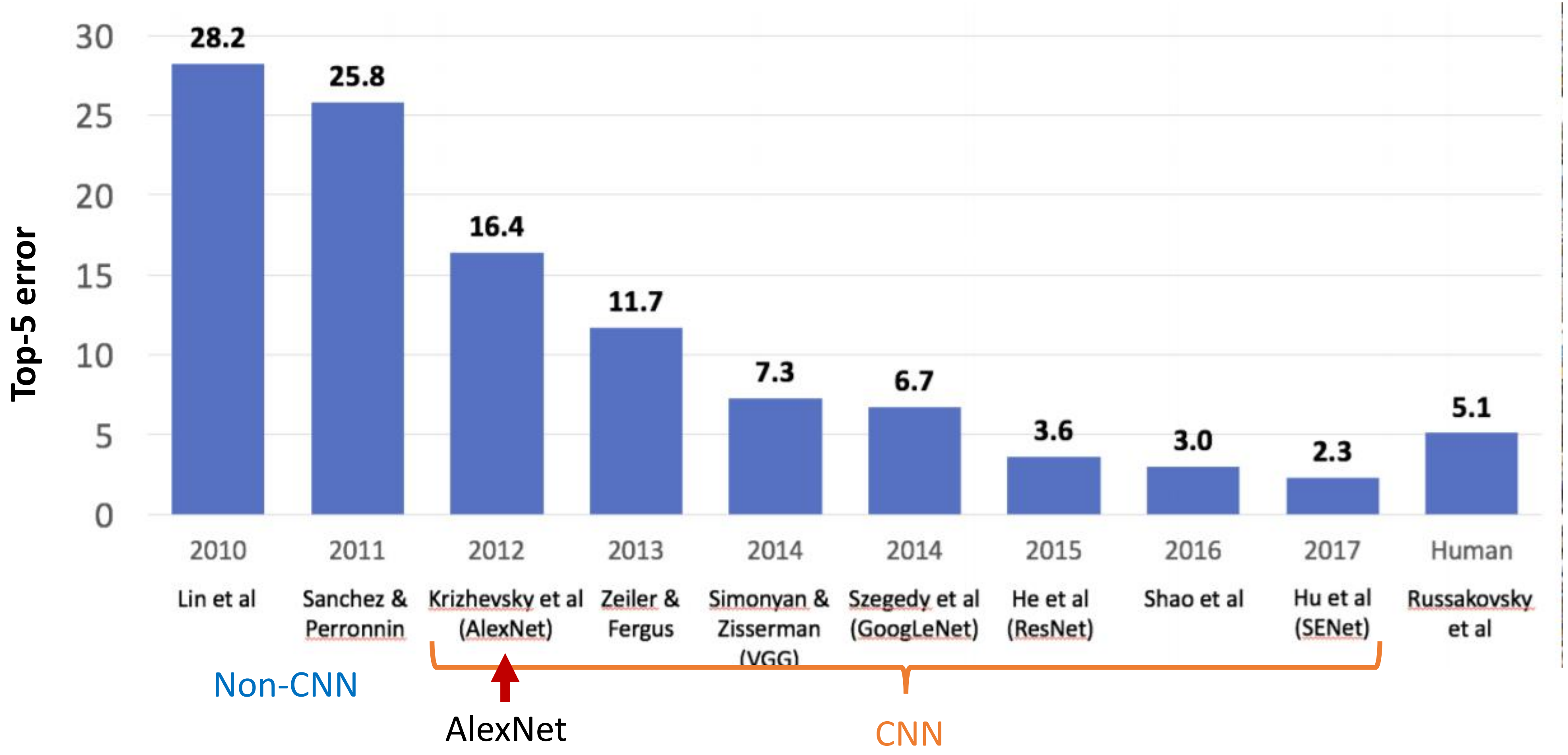
He et al (ResNet)
2016
Shao et al
2017
Hu et al (SENet)
Human
Russakovsky et al
Non-CNN
CNN

AlexNet



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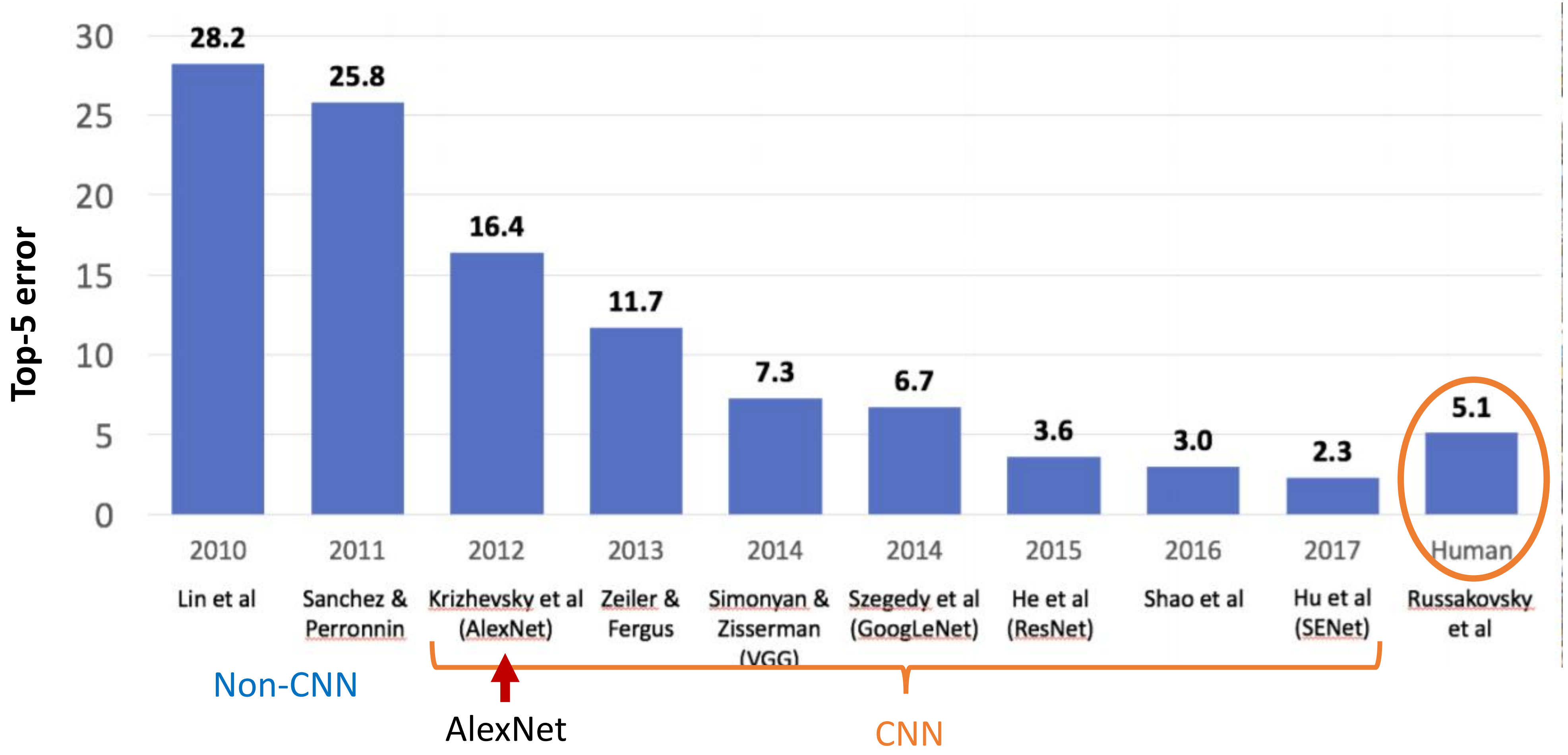
AlexNet



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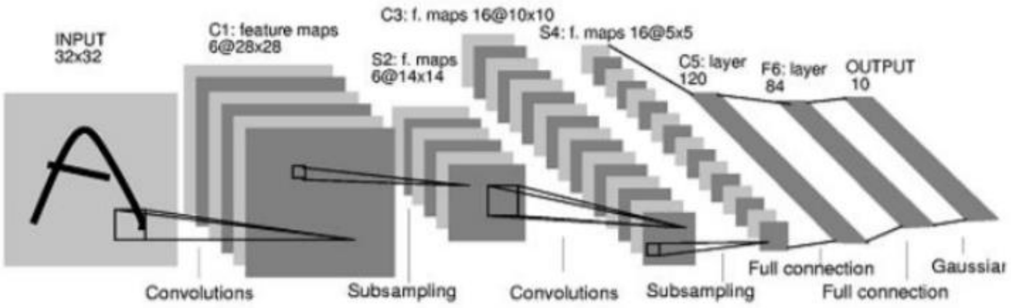
Russakovsky et al
Non-CNN
AlexNet
CNN

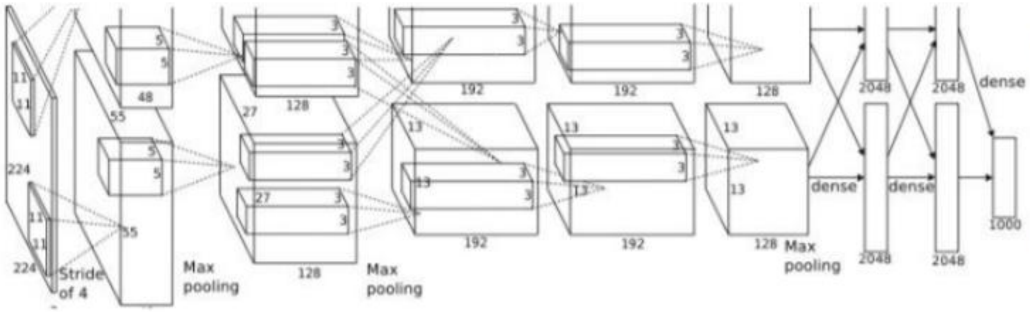
What has changed?

1998
LeCun
et al.



- MNIST digit recognition dataset
- 10^7 pixels used in training

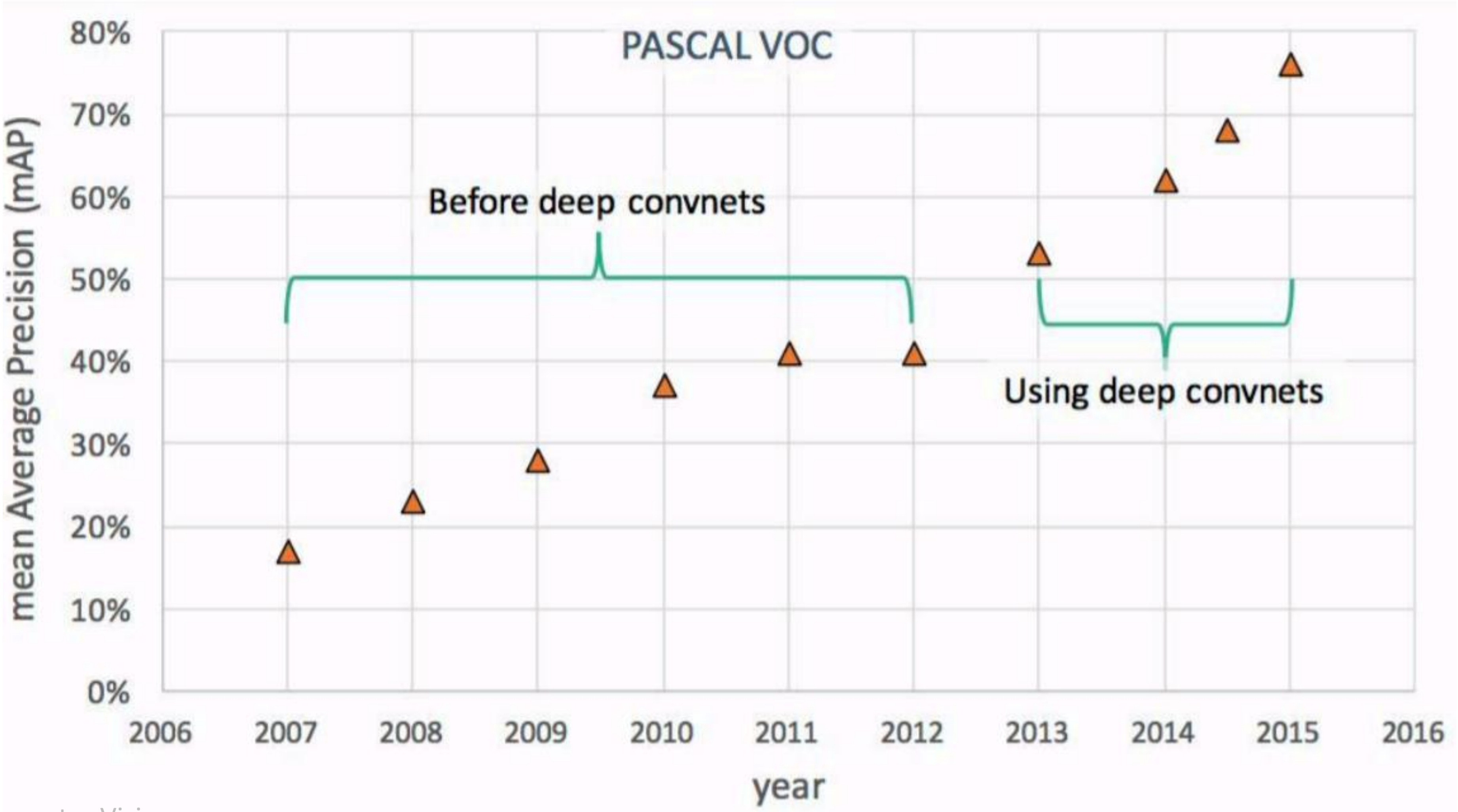
2012
Krizhevsky
et al.



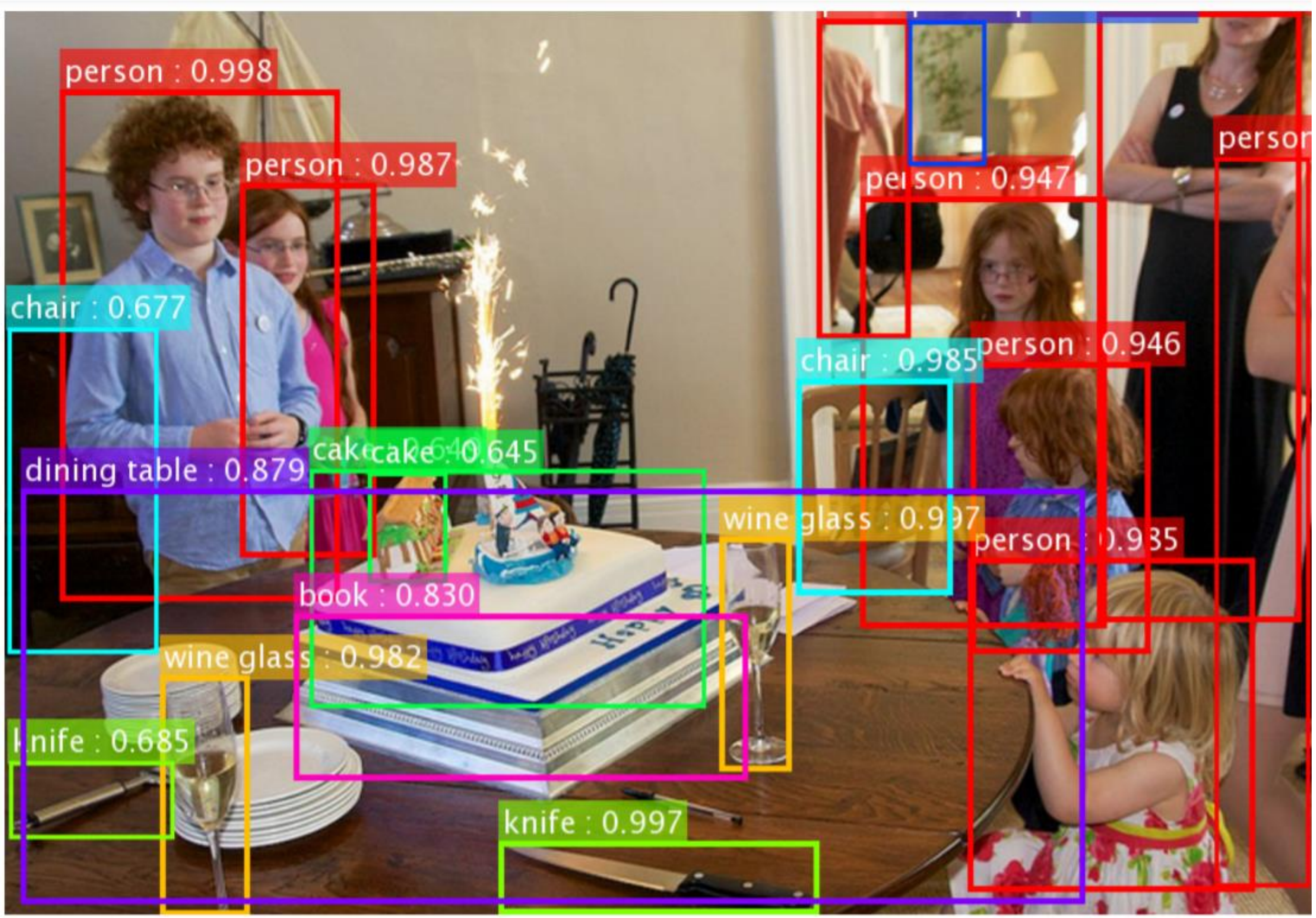
- ImageNet image recognition dataset
- 10^{14} pixels used in training

Source: Niessner and Leal-Taixé, Introduction to Deep Learning

Object Detection Progress

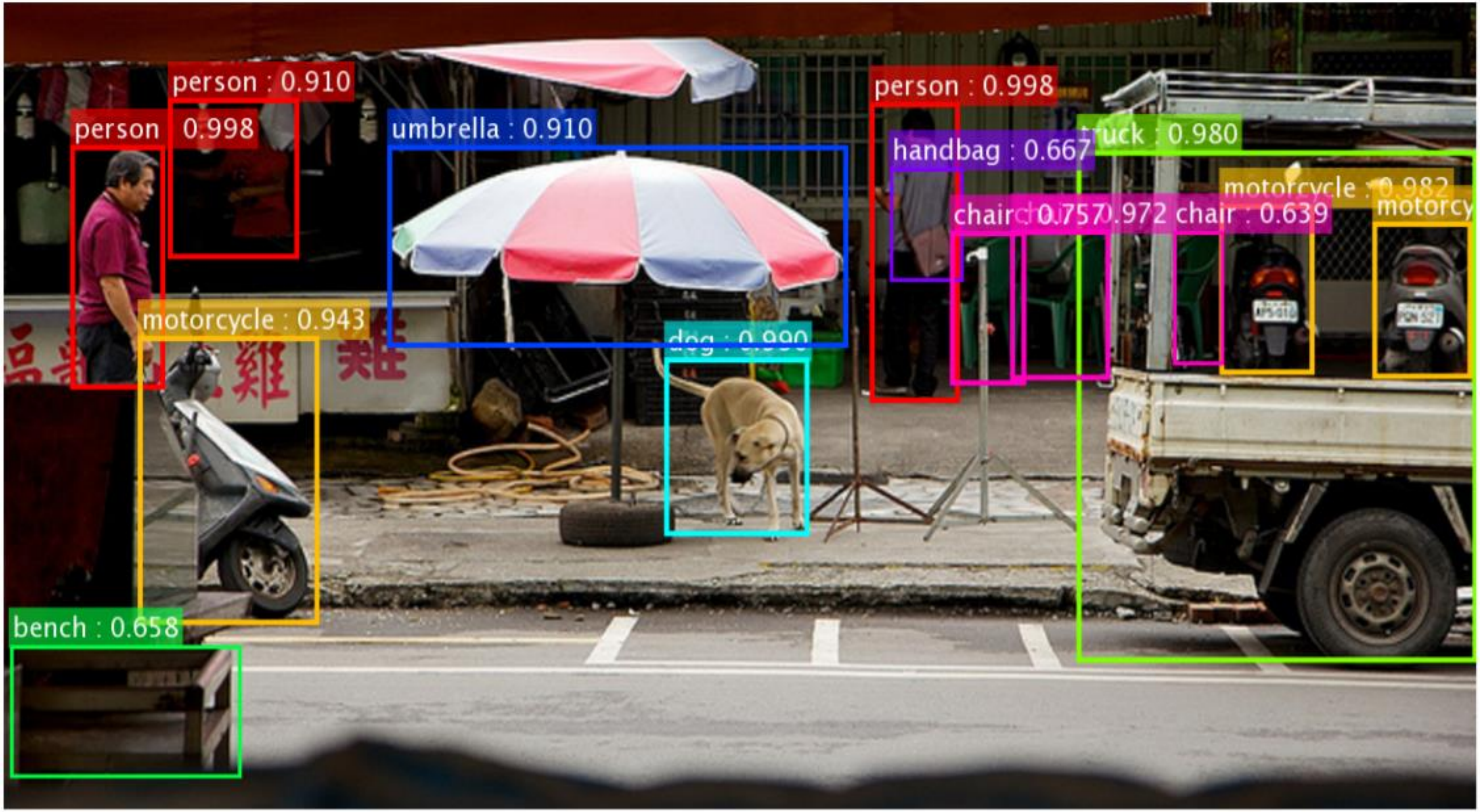


Source: R. Fergus, Computer Vision



Source: R. Fergus, Computer Vision

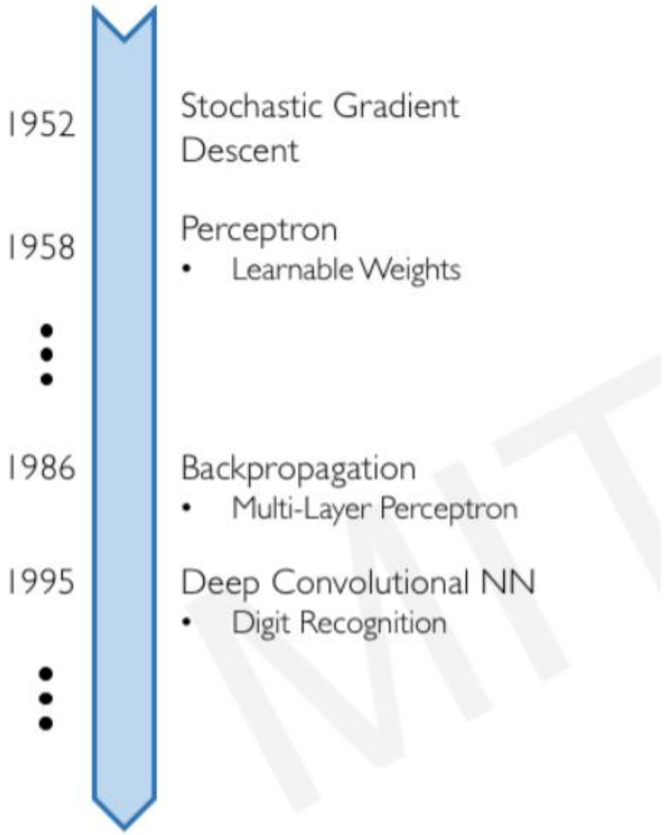
He, Zhang, Ren, & Sun. "Deep Residual Learning for Image



Source: R. Fergus, Computer Vision

He, Zhang, Ren, & Sun. “Deep Residual Learning for Image

Why now?



Source: Amini and Soleimany, MIT 6.S191: Introduction to Deep Learning

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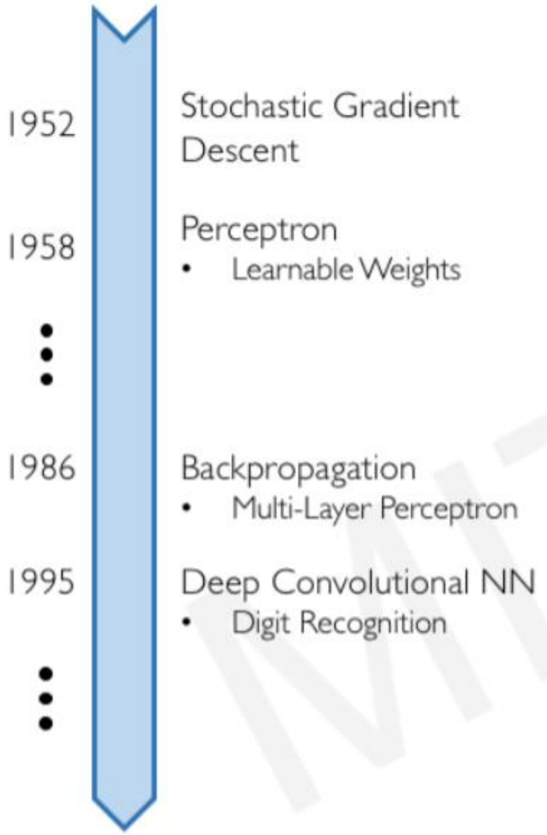
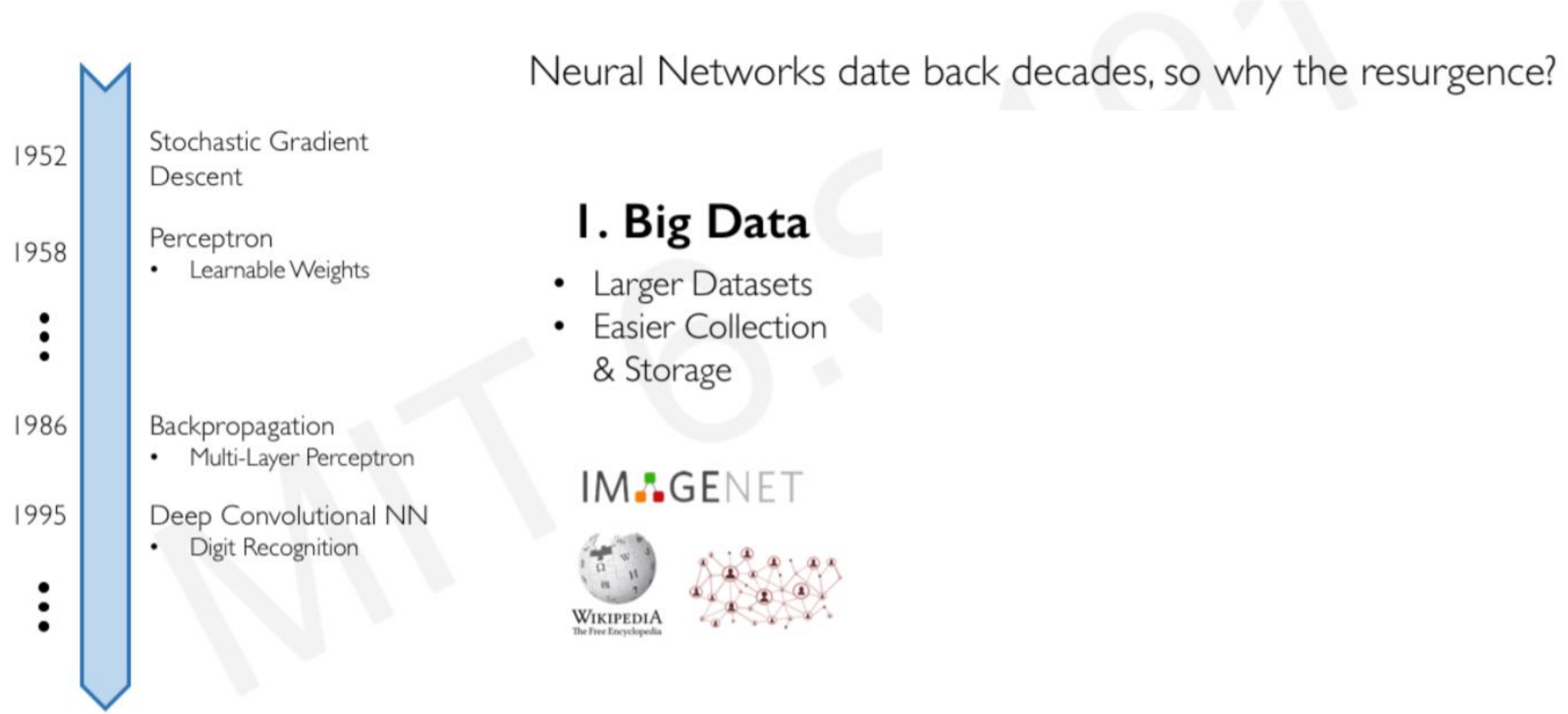


Neural Networks date back decades, so why the resurgence?

Source: Amini and Soleimany, MIT 6.S191: Introduction to Deep Learning

Why now?



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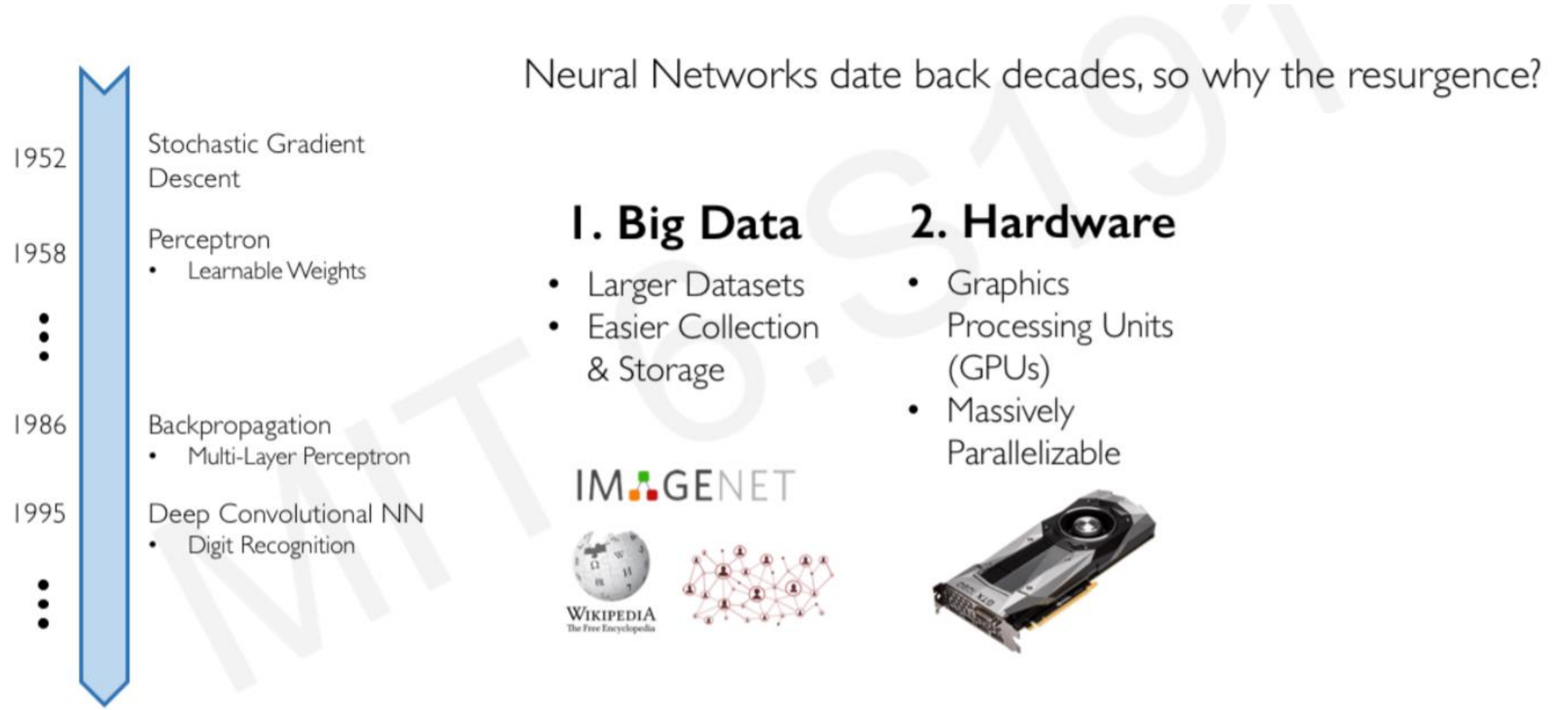

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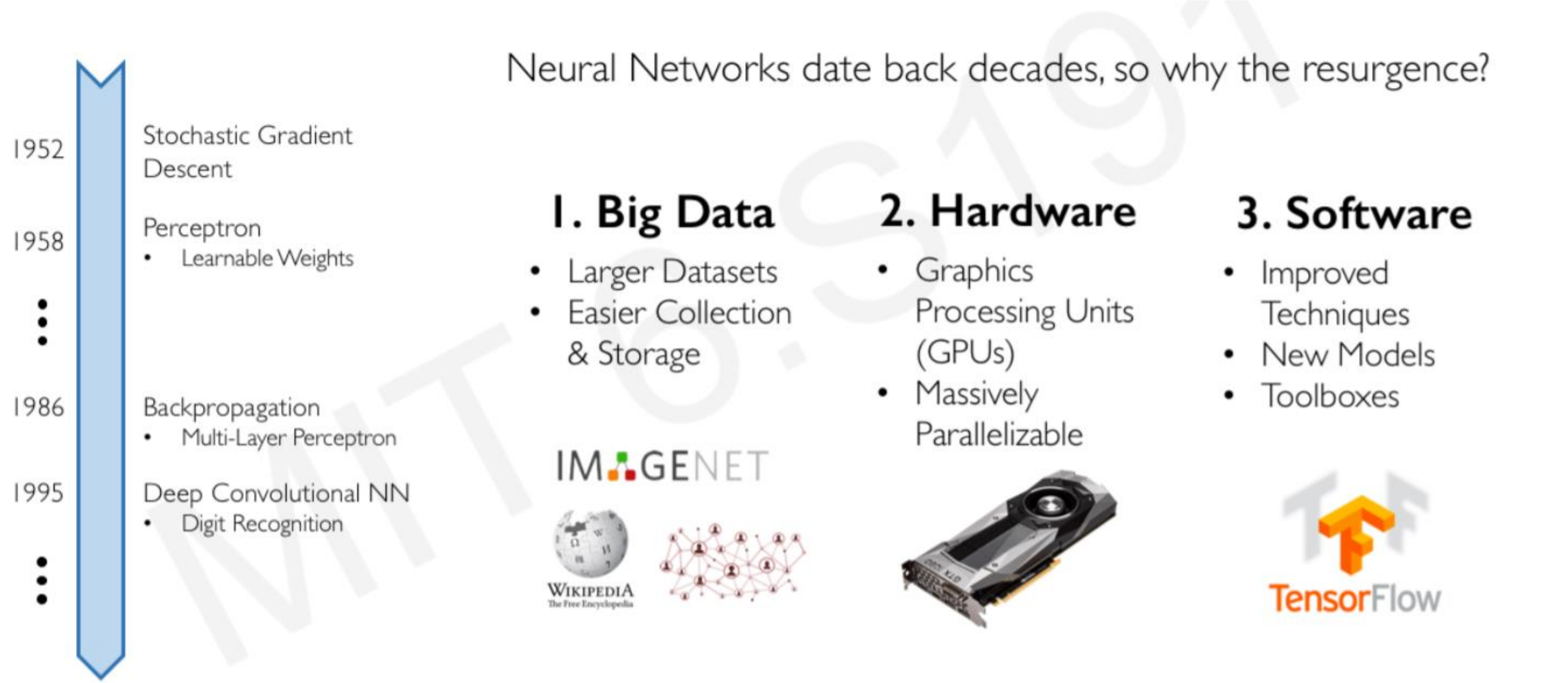


Source: Amini and Soleimany, MIT 6.S191: Introduction to Deep Learning

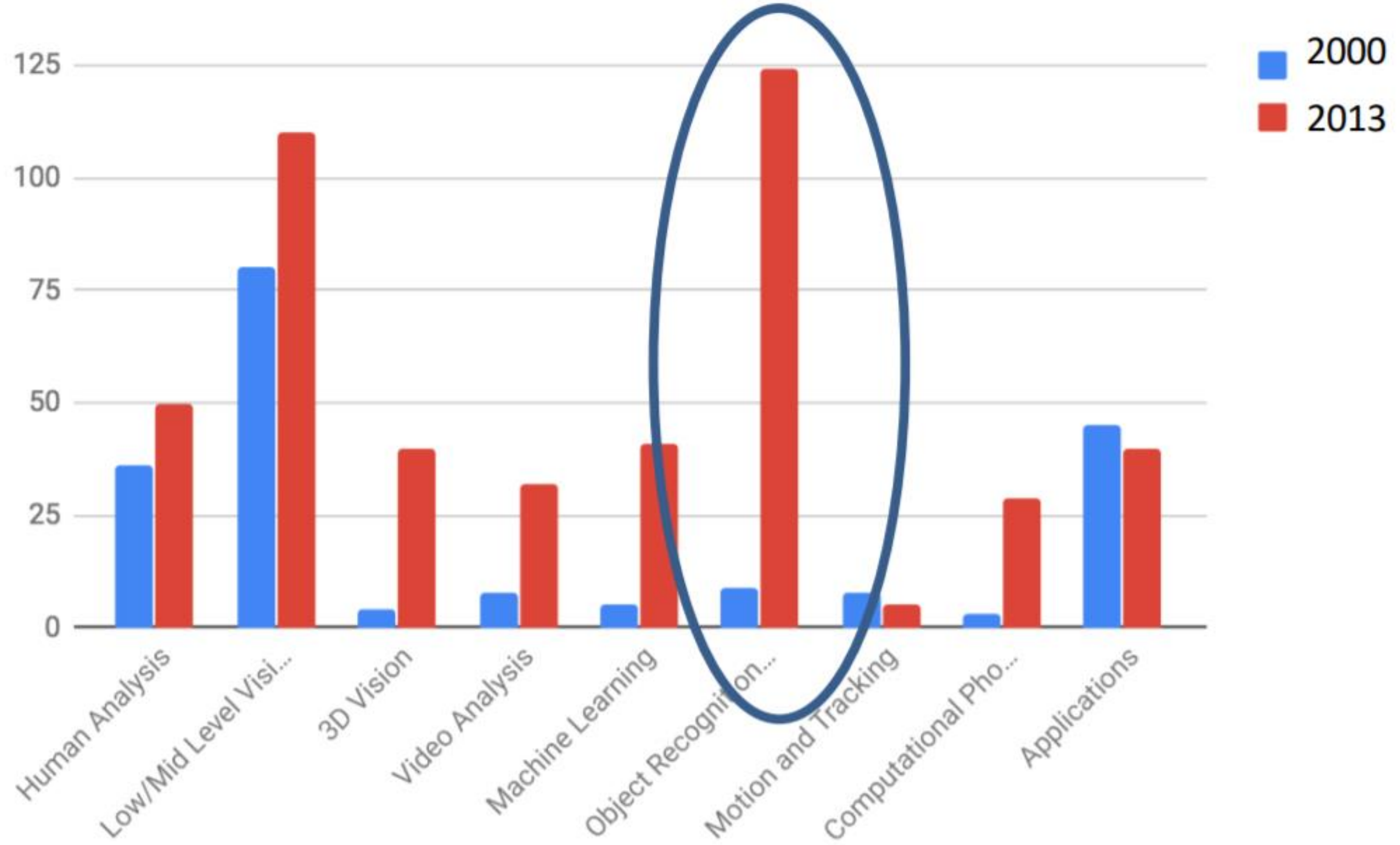
Deep Learning Recognition

Yann LeCun Geoffrey Hinton Yoshua Bengio

ACM Turing Award 2018 (Nobel Prize of Computing)

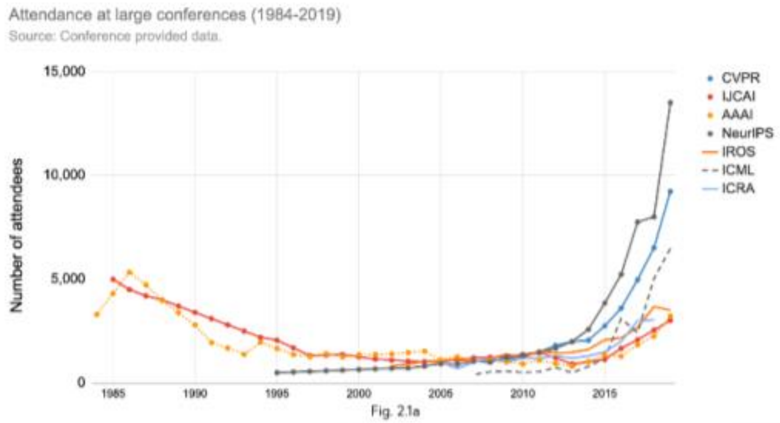
https://en.wikipedia.org/wiki/Turing_Award

Deep Learning and Computer Vision (CVPR topic distribution: 2000 vs. 2013)



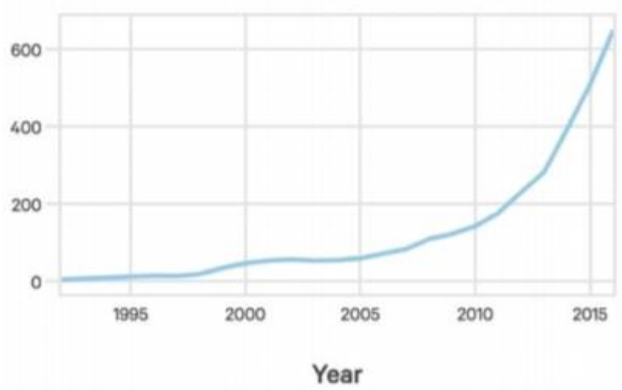
Source: Fei-Fei Li, Ranjay Krishna, Danfei Xu, CS231n: Convolutional Neural Networks for Visual Recognition

AI’s Explosive Growth & Impact



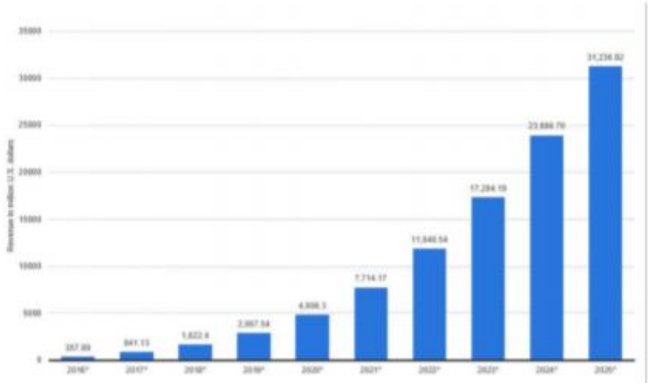
Number of attendance At AI conferences

Source: The Gradient



Startups Developing AI Systems

Source: Crunchbase, VentureSource, Sand Hill Econometrics



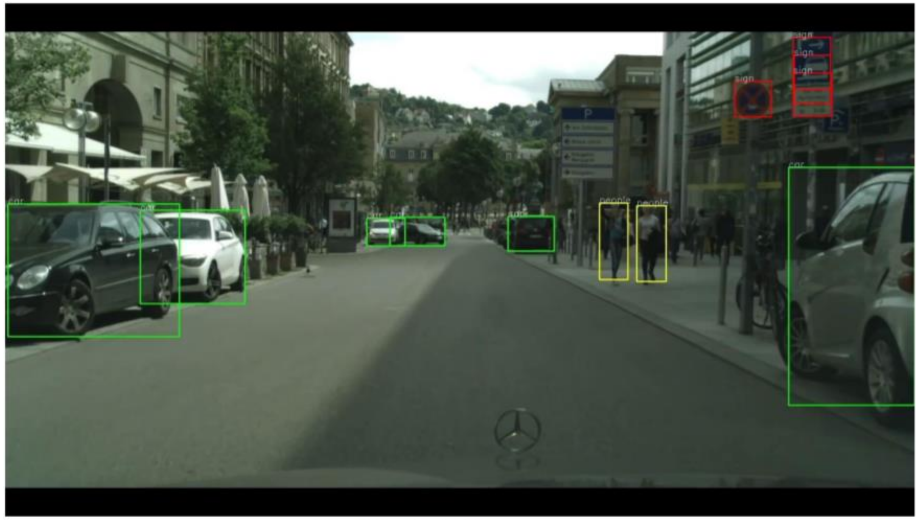
Enterprise Application AI Revenue

Source: Statista

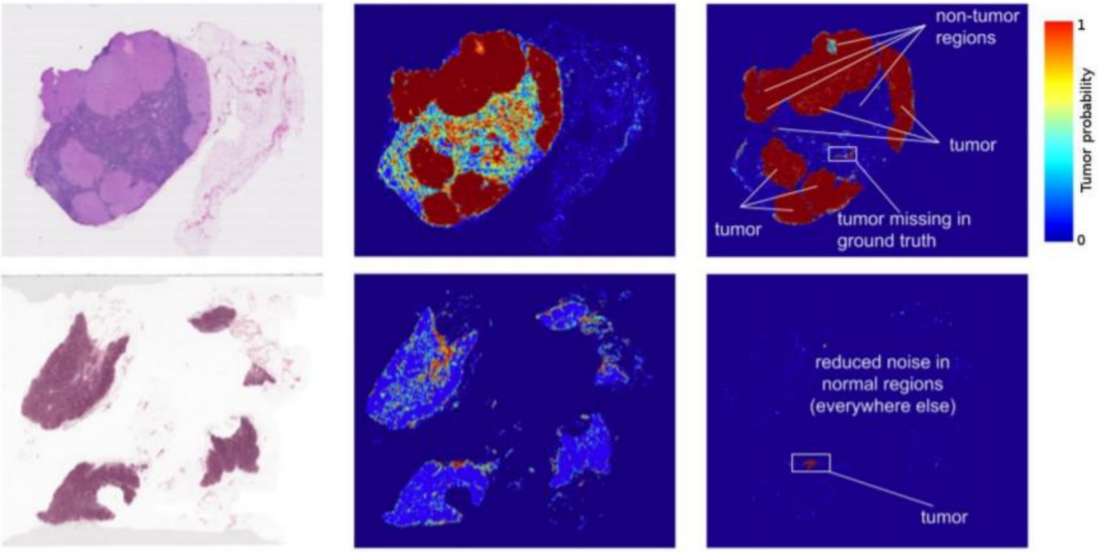
Deep Learning Today

Object Detection

Speech recognition
Machine translation
Automated Text Generation
AlphaGo
AlphaStar
Google Assistant,
etc,



Self-driving cars



Healthcare, cancer detection

Source: Niessner and Leal-Taixé, Introduction to Deep Learning